The Future of Growth and Human Labor Under Recursive AI Self-Improvement

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Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI production services and uses AI research services to improve AI efficiency, generating recursive self-improvement. I characterize the model's long-run equilibria under a finite upper bound on AI efficiency. The equilibrium regime depends jointly on the elasticity of substitution between labor and AI and on the level of this upper bound. When labor remains a production bottleneck, output per person and real wages grow at an exogenously given rate of labor-augmenting technological progress, and labor retains a positive income share. If the elasticity of substitution exceeds one and the upper bound is sufficiently high, the economy instead becomes AI-dominated. In this regime, real-wage growth exceeds the rate of labor-augmenting technological progress, but output per person grows even faster. Labor's income share therefore converges to zero. Simulations show that output growth can remain moderate for a long time before accelerating sharply as AI production services expand.

1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity. One fear is that workers who depend on labor income could lose relative economic power. This fear, however, does not require real wages to decline: workers may receive a smaller share of a much larger economy even as their wages rise. Can unprecedented prosperity coexist with a vanishing labor income share? Under what conditions could the world economy follow such a path? How long could it take before the transition became visible, and what economic signals would reveal it? What would happen to real wages and labor's income share along the way?

I study these questions by extending the Ramsey–Cass–Koopmans model. Competitive firms produce the final good with capital, effective labor, and AI production services. A monopolistic AI developer supplies these services and uses AI research services to improve AI efficiency, creating recursive self-improvement. Together, the elasticity of substitution between AI production services and labor and the maximum attainable level of AI efficiency determine whether human labor remains a long-run production bottleneck.

The escape from the bottleneck arises from two feedback loops. A technological loop runs from AI efficiency to AI research: greater efficiency produces more effective research services from a given amount of research compute and thereby helps create still better AI. An economic loop runs from

AI efficiency to output and back to AI research: better AI raises current production and expands the resources that can finance research compute and therefore further AI improvements. Research compute, however, competes with the inference compute used in final-good production.

I first characterize the labor-bottleneck regime. It arises when the elasticity of substitution between AI production services and labor is at most one, and can also arise when the elasticity exceeds one but the maximum attainable level of AI efficiency remains below an explicit threshold. Along equilibrium paths that converge to this regime, output per person and the real wage eventually grow at the exogenous rate of labor-augmenting technological progress, and labor's income share converges to a positive value.

The outcome changes when the elasticity of substitution between AI production services and labor exceeds one and the maximum attainable level of AI efficiency lies above the threshold. An AI-dominated long-run regime then becomes possible. Along equilibrium paths that converge to this regime, capital and effective AI production services eventually grow faster than effective labor. The long-run growth rates of output per person and the real wage exceed their labor-bottleneck values, and the interest rate is higher. Because output per person grows faster than the real wage, labor's income share converges to zero. AI-driven output growth raises wages despite the shrinking fraction of output paid to labor. Within the AI-dominated regime, long-run growth and the interest rate increase with the maximum attainable level of AI efficiency.

Simulations show that the AI-dominated long-run regime need not become visible quickly. The economy may remain on a moderate-growth path for a long time. Rapid growth of effective AI production services and an interest rate above the labor-bottleneck benchmark can appear before the output and wages start to accelerate. When the economy begins far from its steady state, the transition could also require a large initial research outlay relative to AI-industry revenue and temporary developer losses that are offset in present value by future monopoly rents. Alternative calibrations produce fast or slow transitions.

This paper has six sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines equilibrium. Section 4 establishes sufficient conditions for equilibrium in each substitution regime and characterizes the associated long-run growth and distributional outcomes. Section 5 presents some equilibrium numerical simulations. Section 6 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

AI, growth, and automated research. The research technology builds on the endogenous-growth tradition, in which resources devoted to nonrival knowledge determine long-run growth (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995). Recent work asks how AI affects the economy when AI can automate production, research, or both. Trammell and Korinek (2023) distinguish the automation of ordinary production from the automation of R&D. Mullainathan and Rambachan

(2025) argue that AI may also reorganize the scientific process by bringing idea generation, theory selection, and other research activities that currently occur off screen into algorithmic workflows. The closest feedback architecture is Davidson et al. (2026): more capable AI makes subsequent AI research more productive, while higher output expands the resources that can finance that research. Korinek and Suh (2024) study how an exogenous automation frontier interacts with capital accumulation to shape output and wages during a possible transition to AGI. Korinek et al. (2026) take a complementary scenario-based approach. They map alternative paths of AI capabilities and adoption into GDP, factor shares, wages, occupational reallocation, and unemployment through 2030, but embed recursive improvement in the assumed path of AI task productivity. I instead place the technological and economic feedbacks inside a decentralized Ramsey economy in which consumption, saving, investment, compute, and prices are equilibrium outcomes. This also differs from quantitative scenario models such as Erdil et al. (2025), which combine AI development, automation, investment, and growth but abstract from the decentralized price system and market structure studied here.

Substitution, bottlenecks, and long-run growth. Following Davidson et al. (2026), I call an input or task a bottleneck when its slow improvement constrains aggregate progress even if other inputs or tasks improve rapidly. Aghion et al. (2019) apply Baumol's cost-disease logic to AI and growth: if some essential production tasks remain difficult to automate, progress in automatable tasks need not translate into comparably rapid aggregate growth. Jones (2025) develops the corresponding mechanism inside R&D. The effect of AI on research depends jointly on the share of research tasks AI can perform, its productivity in those tasks, and the strength of complementarities across tasks. Tasks that AI cannot perform can therefore limit overall research progress. In my model, human labor no longer enters AI research, but it may remain essential in final-good production.

The production side connects this bottleneck logic to the classic result that sufficient substitution toward accumulable inputs can prevent their marginal product from vanishing and produce an asymptotically linear technology (Pitchford, 1960; Jones and Manuelli, 1990; Rebelo, 1991). Duffy and Papageorgiou (2000) emphasize that a high elasticity is not by itself sufficient: the productivity of the accumulable input must also be high enough. In the AI literature, Nordhaus (2021) identifies substitutability between information and conventional inputs as central to an economic singularity. Jones and Tonetti (2026) show instead how a small number of weak links can delay growth acceleration even when automation advances. Consistent with this mechanism, Demirer et al. (2026) find that large AI-induced gains in coding activity attenuate sharply at the level of completed software projects and releases, and estimate strong complementarity between AI and human effort. Finally, Restrepo (2025) distinguishes work that remains a bottleneck from work that is no longer essential for growth. I obtain a related distinction without assuming that all work can be performed by AI. Gross substitution between AI services and labor is necessary but not sufficient for labor to cease being a bottleneck: attainable AI efficiency must also be high enough. I derive the threshold separating the labor-bottleneck and AI-dominated equilibria. The capability threshold in Gans (2025) concerns a different mechanism—whether AI directs research toward dense or exploratory regions of the knowledge space—rather than whether effective AI services can replace labor in final production.

Wages and the labor income share. The distributional results build on the task approach to automation. Acemoglu and Restrepo (2019) distinguish the displacement effect of automating existing tasks from two forces that can increase labor demand: the expansion of output generated by higher productivity and the creation of new tasks in which labor retains a comparative advantage, which they call the reinstatement effect. My model does not create new human tasks, so it does not contain this reinstatement margin. It captures the displacement and productivity forces. Labor and AI services enter a constant-elasticity-of-substitution (CES) production function. Holding capital and effective labor fixed, greater use of AI services raises labor's income share when the elasticity is below one, leaves it unchanged when the elasticity equals one, and reduces it when the elasticity exceeds one. This mechanism is central to the analysis of the declining labor share in Karabarbounis and Neiman (2014). Recent work shows that automation can raise wages while reducing labor's income share (Autor and Kausik, 2026). In the long run, Ray and Mookherjee (2022) show that capital accumulation can drive labor's share to zero while real wages rise indefinitely, and Restrepo (2025) obtains a vanishing labor share when compute can perform all bottleneck work. The coexistence of rising real wages and a declining labor share is therefore not itself new. Farboodi et al. (2026) obtain a contrasting wage outcome: with endogenous capital accumulation, data-driven automation can generate explosive growth but stagnant long-run wages. My contribution is a sharper equilibrium characterization. Among the long-run regimes characterized here, real-wage growth exceeds the exogenous rate only in the AI-dominated regime, in which labor's income share converges to zero. In every positive-labor-share regime, real-wage growth equals the exogenous rate. Within the AI-dominated regime, an exact expression links the wage-growth premium to the rate at which labor's income share declines, while the AI-efficiency threshold determines which regime arises. The possible erosion of labor and human-consumption tax bases motivates the public-finance analysis in Korinek and Lockwood (2026). I do not study taxation, but the regime results identify when that policy problem can arise.

The aggregation of labor also matters for interpreting these results. Korinek et al. (2026) show that average wages can rise while wages for cognitive workers fall and their unemployment increases. My representative-worker model cannot characterize this heterogeneity within labor. It instead isolates the conditions under which aggregate real wages rise faster than labor-augmenting technological progress while labor's income share converges to zero.

Interest rates. AI-driven growth can also affect equilibrium returns. Faster expected consumption growth raises long-term real rates through consumption smoothing in Chow et al. (2026), and the AI scenarios in Rachel (2025) and Melina and Villa (2025) can raise the natural rate through different saving and investment channels. Using recent AI-sector investment, Wachter and Wachter

(2026) infer scenarios with rapid output and AI-sector expansion and a higher risk-free rate. The sign is not universal in richer models. I study a narrower object: the net return on physical capital in a closed-economy Ramsey equilibrium. The return remains tied to labor-augmenting technological progress when labor is the bottleneck, but rises with attainable AI efficiency in the AI-dominated regime.

Measurement of AI services. The distinction between raw compute and the effective services it produces is informed by Korinek and McKelvey (2026), who distinguish physical computing capacity from quality-adjusted inference and R&D/training output and account for algorithmic progress. I retain separate compute flows for inference and research but summarize the technology that augments both in one AI-efficiency stock. Coyle and Poquiz (2025) document the broader difficulties of measuring AI investment and output, quality change, and changes in production processes. These measurement problems caution against interpreting the model's AI-efficiency stock as a directly observed statistical series.

Market structure. The integrated developer is a tractable limiting case of the concentration forces studied by Korinek and Vipra (2025) and the AI market-power questions surveyed by Gans (2024). Athey and Scott Morton (2025) model AI as a priced intermediate input and show how upstream market power affects adoption, downstream prices, factor returns, and welfare. Evidence on model pricing, entry, and differentiation in Demirer et al. (2025) also cautions against treating current AI markets as a literal monopoly. I hold market structure fixed to isolate the dynamic mechanism. This accounting distinguishes the AI industry's gross revenue from monopoly profit: inference and research expenditure are paid before the developer distributes the residual to its owner.

Contribution. The individual mechanisms studied here—automated research, production bottlenecks, declining labor shares, rising real wages, and changes in equilibrium returns—appear in the existing literature. I combine them in a decentralized endogenous-growth model with recursive AI improvement and a profit-maximizing AI developer. Across the long-run equilibria characterized here, real wages grow faster than exogenous labor-augmenting technological progress if and only if labor's income share converges to zero. Output per person then grows even faster, and an exact expression links the wage-growth premium to the rate at which labor's income share declines. Within the AI-dominated regime, higher attainable AI efficiency raises both long-run growth and the net return on capital.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which the replacement of human researchers at the technological frontier has already occurred.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \qquad n \geq 0. \tag{1}$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N. Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL. Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \qquad \gamma > 0. \tag{2}$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the net distribution the household receives from the AI developer. The household takes the paths of the net interest rate r, the wage w and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t\geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t)\,\mathrm{d}t, \qquad \rho > n, \tag{3}$$

subject to

$$\dot{K} = rK + wN + \Pi - C. \tag{4}$$

Admissible choices satisfy $C_t > 0$, $K_t \geq 0$, and $K(0) = K_0$. The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \tag{5}$$

Its transversality condition is

$$\lim_{t\to\infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \tag{6}$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K, effective labor AL, and effective AI production services X. Let $0 < \alpha < 1$ denote the capital elasticity, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services.

The CES weights ω_L and ω_X are strictly positive and sum to one. Production is

$$Y = K^{\alpha} Z^{1-\alpha}, \tag{7}$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)}. \tag{8}$$

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others (Autor, 2013; Acemoglu and Restrepo, 2019). I abstract from differences across workers, occupations, and tasks by modeling homogeneous labor and aggregate effective AI production services. The elasticity of substitution governs the relationship between these aggregate inputs, rather than the allocation of individual tasks to humans or AI. This simplification focuses the analysis on aggregate wages and labor's income share, leaving differences in exposure and earnings across workers outside the model.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta, w$, and p_X as given, a final-good firm solves

$$\max_{K,L,X \geq 0} \left\{ K^{\alpha} Z(AL, X)^{1-\alpha} - (r+\delta)K - wL - p_X X \right\}. \tag{9}$$

Define s_X as the elasticity of Z with respect to X:

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \tag{10}$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \tag{11}$$

$$w = (1-\alpha)(1-s_X)\frac{Y}{L}, \tag{12}$$

$$p_X = (1-\alpha) s_X \frac{Y}{X}. \tag{13}$$

Constant returns and these conditions imply

$$Y = (r+\delta)K + wL + p_X X. \tag{14}$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The inverse absolute elasticity is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1-s_X}{\sigma} + \alpha s_X. \tag{15}$$

This demand schedule is an input to the developer's problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used to execute the numerical operations required by AI systems (Sastry et al., 2024). The variables U and M are rates of compute use rather than stocks of processors or data centers. I measure compute in units such that its price in final goods is one. Compute is therefore a rival, costly input, not a measure of AI performance.

I distinguish compute used to deploy AI from compute used to improve it. Inference compute U runs existing AI systems and produces outputs used in current final-good production. Research compute M is allocated to the development of better AI. It includes training and fine-tuning, but also computational experimentation, model evaluation, and other compute-intensive components of AI research. I therefore use the term *research compute* rather than the narrower term *training compute*. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training.

Raw compute does not by itself determine the quantity of AI services produced. The variable B measures AI efficiency: AI services obtained per unit of compute. Thus U and M are raw computational inputs, while BU and BM are the corresponding flows of AI production and AI research services. This interpretation parallels measures of algorithmic efficiency based on the decline in compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022). A rise in B therefore represents an improvement in the algorithms that use compute, rather than an expansion of physical computing capacity. I assume that the same B augments compute in both activities.

Inference compute produces AI production services according to

$$X = BU. \tag{16}$$

Equation (16) maps inference compute into the flow $X = BU$ of AI production services. Research compute analogously produces BM, the flow of AI research services.

Figure 1 summarizes the production block.

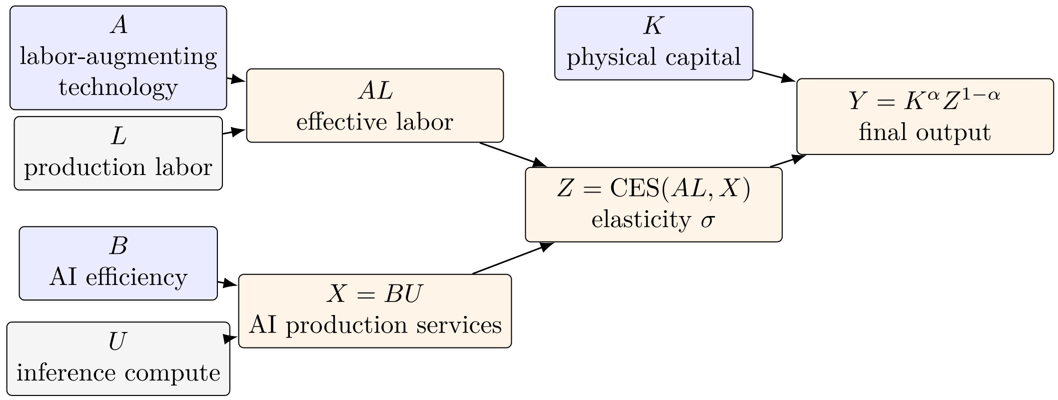

Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z, which combines with capital K to produce Y.

3.4.2 Autonomous improvement in AI efficiency

Given initial AI efficiency $B_0 > 0$ and an efficiency frontier $\overline{B} \in (B_0, \infty]$, AI efficiency evolves according to

$$\dot{B} = \chi (BM)^{\eta} \psi(B), \qquad \psi(B) = 1 - \frac{B}{\overline{B}}, \qquad \chi > 0, \quad 0 < \eta < 1. \tag{17}$$

Figure 2 summarizes the AI-efficiency improvement mechanism.

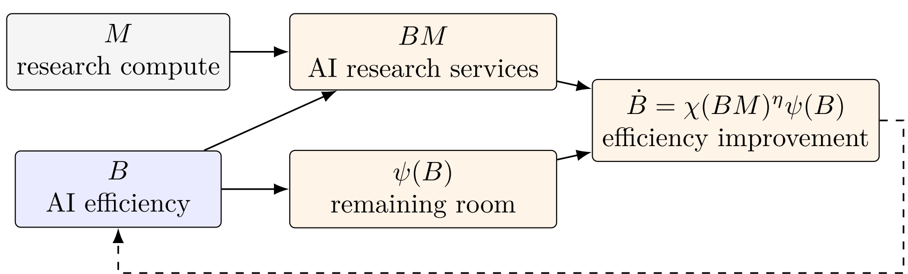

Figure 2: Autonomous improvement in AI efficiency. AI efficiency B augments research compute M to produce BM, the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B.

The parameter χ shifts research productivity. The parameter η governs returns to current research scale. The same AI-efficiency index B augments both inference and research compute. This common-productivity assumption is parsimonious but substantive. Because BM enters the law of motion for AI efficiency, a higher B makes each unit of research compute M more productive and creates recursive improvement. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\overline{B}$. The uncapped specification is recovered at $\overline{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially while research productivity has fallen across several industries, products, and firms. Following this interpretation, Davidson et al. (2026) use their estimates to discipline diminishing returns in idea production. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\overline{B}$ is a deliberately stronger reduced-form specification to make sure there is an equilibrium when AI and labor are substitutes.

The efficiency frontier $\overline{B}$ allows me to establish and compute equilibria for finite values of $\overline{B}$ and then ask whether they have a meaningful limit as $\overline{B} \to \infty$. Without the frontier, recursive improvement may instead produce a finite-time singularity: one or more endogenous quantities—such as AI efficiency, output, capital, or research compute—become unbounded at a finite date, so the trajectory cannot be continued as a path of finite allocations.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B. It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the law of motion for AI efficiency defined by Equation (17) directly, its within-date choice can be solved conditional on B, K, and L. Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \tag{18}$$

The notation $\pi(B)$ leaves its dependence on current K, A, and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \tag{19}$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \tag{20}$$

At a fixed X, a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M. Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B. Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B. Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. Taking the paths of r, K, A, and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B:

$$\max_{\{M_t\}_{t\geq 0}} \int_0^{\infty} \exp\left\{-\int_0^t r_s \, \mathrm{d}s\right\} [\pi(B_t) - M_t] \, \mathrm{d}t, \tag{21}$$

subject to (17) and B_0 given. Admissible research paths have $M_t \geq 0$ and finite total expenditure on every finite interval. The objective is the present value of net distributions, not the flow Π at a particular date.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B. An interior research choice satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \tag{22}$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta\frac{\dot{B}}{B} - \dot{B}\frac{\psi'(B)}{\psi(B)}. \tag{23}$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta\frac{\dot{B}}{B} + q\dot{B}\frac{\psi'(B)}{\psi(B)} + \dot{q}. \tag{24}$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the gain in research productivity. With a finite frontier, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq.

The developer's transversality condition is

$$\lim_{t\to\infty} \exp\left\{-\int_0^t r_s \,\mathrm{d}s\right\} q_t B_t = 0. \tag{25}$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum, but they do not by themselves establish global optimality. Lemma 2 in Appendix A gives a global verification result for the developer's research path. It uses a monotone transformation of the bounded AI-efficiency state and requires the maximized Hamiltonian to have a global supporting tangent at the candidate state. Global concavity is sufficient for this condition but is not necessary. The equilibrium existence proofs use a concavity test, whereas the quantitative analysis checks the weaker support condition directly whenever global concavity fails.

3.5 Equilibrium

I can now combine the conditions of the three agents. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \tag{26}$$

Definition 1 (Equilibrium)**.** *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t\geq 0} \tag{27}$$

such that for every date, the following conditions hold:

1. The intratemporal allocation and prices satisfy

$$L = N, \tag{28a}$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)}, \tag{28b}$$

$$Y = K^\alpha Z^{1-\alpha}, \tag{28c}$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \tag{28d}$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \tag{28e}$$

$$r = \alpha \frac{Y}{K} - \delta, \tag{28f}$$

$$w = (1-\alpha)(1-s_X)\frac{Y}{L}, \tag{28g}$$

$$p_X = (1-\alpha) s_X \frac{Y}{X}, \tag{28h}$$

$$\frac{1}{B} = p_X (1 - e_X), \tag{28i}$$

$$X = BU, \tag{28j}$$

$$1 = q\chi\eta B (BM)^{\eta-1}\psi(B), \tag{28k}$$

$$\Pi = p_X X - U - M. \tag{28l}$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \tag{29a}$$

$$\dot{B} = \chi (BM)^\eta \psi(B), \tag{29b}$$

$$\frac{\dot{C}}{C} = n + r - \rho, \tag{29c}$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta\frac{\dot{B}}{B} - \dot{B}\frac{\psi'(B)}{\psi(B)}. \tag{29d}$$

3. The two transversality conditions are

$$\lim_{t\to\infty} e^{-\rho t}\frac{N_t K_t}{C_t} = 0, \tag{30a}$$

$$\lim_{t\to\infty} \exp\left\{-\int_0^t r_s \,\mathrm{d}s\right\} q_t B_t = 0. \tag{30b}$$

The exogenous paths are $A_t = A_0 e^{\gamma t}$ and $N_t = N_0 e^{nt}$, and the predetermined stocks satisfy $K(0) = K_0$ and $B(0) = B_0$. At $\sigma = 1$, the CES equations use their continuous limits $Z = (AL)^{\omega_L} X^{\omega_X}$ and $s_X = \omega_X$. The trajectory is feasible for the agents' stated problems, the household and developer attain their optima, and final-good firms maximize profit. For a finite frontier, $B_t < \overline{B}$ at every finite date.

Remark 1 (The economy without AI). As a separate boundary case, set $\omega_X = 0$ and $\omega_L = 1$, so that $Z = AL$. Effective AI production services have no productive use, and the developer has no revenue with which to reward improvements in AI efficiency. Thus $X = U = M = \Pi = 0$; B remains constant and does not affect allocations. With $L = N$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress:

$$\begin{aligned} Y &= K^{\alpha}(AN)^{1-\alpha}, & \dot{K} &= Y - C - \delta K, \\ r &= \alpha Y/K - \delta, & \dot{C}/C &= n + r - \rho. \end{aligned}$$

The household transversality condition remains unchanged. The elasticity σ and the AI research parameters are irrelevant in this reduced economy; the interior AI price and costate conditions must not be imposed at $\omega_X = 0$. In particular, $wN/Y = 1 - \alpha$ and $(r + \delta)K/Y = \alpha$.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r+\delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \tag{31}$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute. It is an accounting consequence of equilibrium, not an additional equilibrium condition.

4 Equilibrium existence with a finite AI-efficiency frontier

In this section, I give sufficient conditions for the existence of an equilibrium in each substitution regime. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high. To identify this threshold, I evaluate the output–capital ratio implied by the static equilibrium conditions as AI efficiency approaches its upper bound and labor's income share tends to zero. I compare this ratio with the one required to sustain balanced growth at the exogenous rate of labor-augmenting technological change. If the former exceeds the latter, the resulting return to capital supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

A finite AI-efficiency frontier does not necessarily make long-run growth of output per person equal to γ. It bounds AI efficiency, but not the compute used to supply effective AI production services. If AI and labor are substitutes, capital and compute can sustain growth faster than effective labor even when AI efficiency converges to a finite limit.

To identify the output–capital ratio when labor's income share vanishes, fix $B > 0$ and consider $s_X \to 1$, with $\sigma \neq 1$. Since $wL/Y = (1-\alpha)(1-s_X)$, this limit corresponds to a vanishing labor income share.

First, combining the final-good firm's demand for AI services, equation (13), with the developer's pricing condition, equation (19), gives

$$\frac{X}{Y} = B(1-\alpha)s_X(1-e_X).$$

Equation (15) implies that $e_X \to \alpha$ as $s_X \to 1$. Therefore,

$$\frac{X}{Y} \longrightarrow B(1-\alpha)^2.$$

One factor $1-\alpha$ comes from the marginal product of AI services; the other comes from the developer's marginal-revenue condition.

Second, the CES aggregator and the definition of s_X, equations (8) and (10), imply

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X}\right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}.$$

Combining these two limits yields

$$\frac{Z}{Y} = \frac{Z}{X}\frac{X}{Y} \longrightarrow B(1-\alpha)^2\omega_X^{\sigma/(\sigma-1)}.$$

Finally, dividing the production function, equation (7), by Y and rearranging gives

$$1 = \left(\frac{K}{Y}\right)^{\alpha}\left(\frac{Z}{Y}\right)^{1-\alpha}, \qquad \frac{Y}{K} = \left(\frac{Z}{Y}\right)^{(1-\alpha)/\alpha}.$$

I therefore denote the limiting output–capital ratio by

$$\zeta(B) \equiv \left[(1-\alpha)^2 B\,\omega_X^{\sigma/(\sigma-1)}\right]^{(1-\alpha)/\alpha}. \tag{32}$$

For $\sigma > 1$, $s_X \to 1$ means that AI production services become arbitrarily large relative to effective labor. If AI efficiency also converges to its frontier, then $Y/K \to \zeta(\overline{B})$. I write $\overline{\zeta} \equiv \zeta(\overline{B})$. This calculation identifies a possible limiting output–capital ratio; it does not establish the existence of an equilibrium trajectory approaching it. The propositions below address that question.

Define the threshold $\mathcal{B}(\sigma)$ by

$$\begin{aligned} \zeta(\mathcal{B}(\sigma)) &= \frac{\rho+\gamma+\delta}{\alpha}, \\ \mathcal{B}(\sigma) &= \frac{[(\rho+\gamma+\delta)/\alpha]^{\alpha/(1-\alpha)}}{(1-\alpha)^2\omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \tag{33}$$

The right-hand side of the first line is the output–capital ratio required for output per person to grow at rate γ. Aggregate capital and consumption then grow at $n+\gamma$, so the Euler equation requires $r=\rho+\gamma$ and the capital first-order condition requires $Y/K=(\rho+\gamma+\delta)/\alpha$. The second line gives the threshold explicitly. Appendix A derives these expressions and proves the following propositions. I use $\mathcal{B}(\sigma)$ as an economic regime threshold only when $\sigma>1$. For $\sigma<1$, it appears only as a sufficient condition for the particular local finite-frontier construction below.

For any positive variable V, write $g_V=\dot{V}/V$.

For the next two propositions, I use a normalization that serves both the existence proofs and the numerical algorithm. Capital can grow without bound while the remaining AI-efficiency gap shrinks to zero. I therefore express the initial conditions in coordinates with finite candidate long-run values, which can guide the construction of equilibrium paths. Since $L=N$, effective labor is AN. For the constructions with a positive long-run labor income share, I define

$$k=\frac{K}{AN},\qquad d=(\overline{B}-B)AN,\qquad \tau=\frac{1}{AN}.$$

The variable k is capital per unit of effective labor. Dividing by AN removes the growth of effective labor from the capital stock. The variable d is the remaining AI-efficiency gap scaled by effective labor. Multiplying by AN allows a shrinking gap to have a positive limiting value. In particular,

$$\frac{\dot{d}}{d}=n+\gamma-\frac{\dot{B}}{\overline{B}-B}.$$

A constant d therefore requires the gap to shrink at the rate at which effective labor expands. Finally, $\dot{\tau}=-(n+\gamma)\tau$, so the inverse of effective labor converges to zero. These coordinates leave the model unchanged: at every finite date, $K=k/\tau$ and $B=\overline{B}-d\tau$.

The positive constants $\overline{k},\overline{d}$ introduced in each proposition are the limiting values of k,d along the constructed equilibrium paths. Thus $\overline{k}$ is long-run capital per unit of effective labor, while $\overline{d}$ determines the magnitude of the remaining efficiency gap:

$$\frac{\overline{B}-B(t)}{\overline{d}/[A(t)N(t)]}\longrightarrow 1.$$

The constant $\overline{d}$ is not a convergence rate or an additional parameter. Both constants are determined by the equations of the model for the parameters of the corresponding regime; their derivation is in Appendix A. The propositions establish convergence to these values rather than assuming it.

The algorithm uses the limiting point and the equilibrium dynamics near it to formulate long-run boundary conditions, rather than imposing arbitrary final values for consumption and the shadow price. These conditions help select the initial C_0 and q_0 for given initial stocks; see Appendix B. The normalization does not itself establish equilibrium: optimality and both transversality conditions must still be verified.

Throughout these two propositions, $0 < \overline{B} < \infty$ and initial conditions satisfy $A_0, N_0, K_0 > 0$ and $0 < B_0 < \overline{B}$. Each proposition specifies its own neighborhood $\mathcal{U}_L$ of $(\overline{k}, \overline{d}, 0)$. Closeness is required in all three coordinates: B_0 close to $\overline{B}$ alone is not sufficient for the local construction. The point with $\tau = 0$ describes the long-run limit, not an admissible initial state at a finite date.

4.1 Complementarity: $\sigma < 1$

Proposition 1 (Equilibrium trajectories with complementary inputs)**.** *If* $0 < \sigma < 1$ *and* $\overline{B} > \mathcal{B}(\sigma)$*, then there are constants* $\overline{k}, \overline{d} > 0$ *and a neighborhood* $\mathcal{U}_L$ *of* $(\overline{k}, \overline{d}, 0)$ *such that, for every set of initial conditions satisfying*

$$(k_0, d_0, \tau_0) \in \mathcal{U}_L, \tag{34}$$

there exists an equilibrium trajectory along which

$$\begin{gathered} k \to \overline{k}, \qquad d \to \overline{d}, \qquad \tau \to 0, \qquad B \to \overline{B}, \\ g_Y - n \to \gamma, \qquad g_w \to \gamma, \qquad r \to \rho + \gamma, \qquad \frac{wL}{Y} \to (1-\alpha)(1-\overline{s}_X) > 0. \end{gathered} \tag{35}$$

Here $\overline{s}_X \equiv \lim_{t\to\infty} s_X(t) \in (0,1)$.

Proof. See Appendix A. □

With complementarity, both effective labor and effective AI production services remain relevant in the long run. Effective AI production services expand through compute even as AI efficiency stagnates, and output per person eventually grows at γ.

4.2 Unit elasticity: $\sigma = 1$

At unit elasticity I use the continuous CES limit $Z = (AL)^{\omega_L} X^{\omega_X}$, so $s_X = \omega_X$. Define $\beta = (1-\alpha)\omega_X$. The monopoly condition gives $U = \beta^2 Y$. In this case, the limiting capital stock per unit of effective labor has the explicit expression

$$\overline{k} = \left[(\beta^2 \overline{B})^{\beta} \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta - 1} \right]^{1/(1-\alpha-\beta)}. \tag{36}$$

The denominator of the exponent is positive because $1 - \alpha - \beta = (1-\alpha)\omega_L > 0$. The corresponding value of $\overline{d}$ is derived in (87) in the proof.

Proposition 2 (Equilibrium trajectories at unit elasticity)**.** *If $\sigma = 1$, then there exist $\overline{d} > 0$ and a neighborhood $\mathcal{U}_L$ of $(\overline{k}, \overline{d}, 0)$, with $\overline{k}$ given by (36), such that, for every set of initial conditions satisfying (34), there exists an equilibrium trajectory along which*

$$\begin{gathered} k \to \overline{k}, \qquad d \to \overline{d}, \qquad \tau \to 0, \qquad B \to \overline{B}, \\ g_Y - n \to \gamma, \qquad g_w \to \gamma, \qquad r \to \rho + \gamma, \qquad \frac{wL}{Y} = (1-\alpha)\omega_L > 0. \end{gathered} \tag{37}$$

The initial values $C_0, q_0 > 0$ are locally unique among the trajectories that remain near the normalized long-run limit and converge to it.

Proof. See Appendix A. □

No additional lower or upper bound on the finite frontier is needed for this local construction. The result describes transitions toward balanced growth, not an assumption that the economy starts on a balanced-growth path.

4.3 Gross substitution: $\sigma > 1$

Proposition 3 (Equilibrium trajectories with substitute inputs)**.** *Under the parameter restrictions in Section 3, fix $0 < \overline{B} < \infty$ and $\sigma > 1$.*

If $\overline{B} < \mathcal{B}(\sigma)$, use the same coordinates k, d, τ as in the positive-labor-share constructions above. There exist $\overline{k}, \overline{d} > 0$ and a neighborhood $\mathcal{U}_L$ of $(\overline{k}, \overline{d}, 0)$ such that every $A_0, N_0, K_0 > 0$ and $0 < B_0 < \overline{B}$ satisfying (34) admits an equilibrium with the limits in (35), for a limiting share $\overline{s}_X = \lim_{t\to\infty} s_{X,t} \in (0,1)$. The constants are given by (84) and (87) in the proof.

If $\overline{B} > \mathcal{B}(\sigma)$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$\overline{r} = \alpha\overline{\zeta} - \delta, \qquad \overline{g} = n + \overline{r} - \rho > n + \gamma. \tag{38}$$

There exist a constant $\overline{d} > 0$ and a neighborhood $\mathcal{U}_A$ of $(0, \overline{d}, 0)$ such that every $A_0, N_0, K_0 > 0$ and $0 < B_0 < \overline{B}$ satisfying

$$\left[\left(\frac{A_0 N_0}{K_0}\right)^{\varphi},\ (\overline{B} - B_0)K_0,\ \frac{1}{A_0 N_0}\right] \in \mathcal{U}_A \tag{39}$$

admits an equilibrium. Here $\overline{d}$ is the limiting value of $(\overline{B} - B)K$, rather than $(\overline{B} - B)AN$, and is given by (90) in the proof. Along this equilibrium, $r \to \overline{r}$, $s_X \to 1$, and

$$B \to \overline{B}, \qquad g_Y - n \to \overline{r} - \rho > \gamma, \qquad \frac{wL}{Y} \to 0, \tag{40}$$

$$g_w \to \gamma + \frac{\overline{r} - \rho - \gamma}{\sigma} > \gamma. \tag{41}$$

In both cases the path satisfies Definition 1, and the initial values $C_0, q_0 > 0$ are locally unique. The equality $\overline{B} = \mathcal{B}(\sigma)$ is not covered.

Proof. See Appendix A. □

Above the threshold, a fixed AI-efficiency level allows output, capital, and inference compute to grow faster than effective labor. Production becomes asymptotically linear in capital, as in an AK model. Growth is finite for every fixed frontier, but need not equal γ. Real wages rise even as labor's income share vanishes. Below the threshold, labor retains a positive income share and long-run growth per person equals γ.

Proposition 4 (Wage growth and labor's income share)**.** *Consider the long-run equilibria constructed in Propositions 1, 2, and 3. For $\sigma > 1$, exclude $\overline{B} = \mathcal{B}(\sigma)$. Then*

$$\lim_{t\to\infty} g_w > \gamma \quad \Longleftrightarrow \quad \lim_{t\to\infty} \frac{wL}{Y} = 0,$$

and either condition holds precisely when the equilibrium is AI-dominated. Along such an equilibrium, output per person grows asymptotically faster than the real wage:

$$\lim_{t\to\infty} \big[(g_Y - n) - g_w\big] = \left(1 - \frac{1}{\sigma}\right)(\overline{r} - \rho - \gamma) > 0. \tag{42}$$

Because $L = N$, the same wedge is the asymptotic rate at which labor's income share falls:

$$\lim_{t\to\infty} \frac{\mathrm{d}}{\mathrm{d}t} \log\left(\frac{wL}{Y}\right) = -\left(1 - \frac{1}{\sigma}\right)(\overline{r} - \rho - \gamma) < 0. \tag{43}$$

Equivalently,

$$-\lim_{t\to\infty} \frac{\mathrm{d}}{\mathrm{d}t} \log\left(\frac{wL}{Y}\right) = (\sigma - 1) \lim_{t\to\infty} (g_w - \gamma) > 0. \tag{44}$$

Proof. See Appendix A. □

The result is not that workers must become poorer, or that a falling labor share causes wages to grow faster. In the AI-dominated regime, output per person grows rapidly enough to raise the real wage despite the shrinking fraction of output paid to labor. Because output per person grows faster than the real wage, labor receives a vanishing share of an expanding economy. The wedge is zero in the positive-labor-share regimes, where both growth rates converge to γ. Thus $\sigma > 1$ alone is insufficient: the frontier must also place the economy in the AI-dominated regime.

4.4 Deriving long-run growth and the effect of the frontier

The constructed trajectories have convergent growth rates and a positive limiting consumption–output ratio. Write $\overline{r} = \lim_{t\to\infty} r_t$. Consumption and output therefore have the same limiting growth rate. The household Euler equation then gives

$$\lim_{t\to\infty} \frac{\mathrm{d}\log(Y_t/N_t)}{\mathrm{d}t} = \lim_{t\to\infty} (g_C - n) = \overline{r} - \rho. \tag{45}$$

In the positive labor-share regimes, $Y/(AN)$ converges to a positive constant and its growth rate converges to zero. Consequently,

$$g_Y \to n + \gamma, \qquad g_Y - n \to \gamma, \qquad \overline{r} = \rho + \gamma. \tag{46}$$

These are limits of the constructed transitions, not restrictions imposed on their growth rates at every date.

In the AI-dominated regime, $s_X \to 1$ and $e_X \to \alpha$. The CES function and the monopoly condition imply

$$\frac{Z}{X} \to \omega_X^{\sigma/(\sigma-1)}, \qquad \frac{U}{Y} \to (1-\alpha)^2, \qquad \frac{X}{Y} \to \overline{B}(1-\alpha)^2. \tag{47}$$

Production gives the identity $Y/K = (Z/Y)^{(1-\alpha)/\alpha}$. Substitution of these limits therefore yields

$$\frac{Y}{K} \to \overline{\zeta}, \qquad \overline{r} = \alpha\overline{\zeta} - \delta, \qquad \lim(g_Y - n) = \alpha\overline{\zeta} - \delta - \rho. \tag{48}$$

The last expression exceeds γ exactly when $\overline{B} > \mathcal{B}(\sigma)$. A finite AI-efficiency level can thus support sustained growth above γ: inference compute expands with output, and effective labor ceases to constrain production asymptotically.

Holding $\sigma > 1$ and all other parameters fixed, the constructed families give the following comparative statics across finite frontiers:

$$\overline{r}(\overline{B};\sigma) = \begin{cases} \rho + \gamma, & 0 < \overline{B} < \mathcal{B}(\sigma), \\ \alpha\overline{\zeta} - \delta, & \overline{B} > \mathcal{B}(\sigma). \end{cases} \tag{49}$$

Below the threshold, a larger frontier does not change the limiting interest or growth rate. Above it,

$$\frac{\partial \overline{r}}{\partial \overline{B}} = \frac{1-\alpha}{\overline{B}}\overline{\zeta} > 0, \qquad \lim_{\overline{B}\to\infty} \overline{r}(\overline{B};\sigma) = \infty. \tag{50}$$

The limiting growth rate per person has the same derivative because it equals $\overline{r} - \rho$. The two interest-rate formulas approach $\rho + \gamma$ at the threshold, although the existence proof does not cover equality. For $\sigma = 1$, and for the constructed complementary-input family, $\overline{r} = \rho + \gamma$ instead remains independent of the frontier.

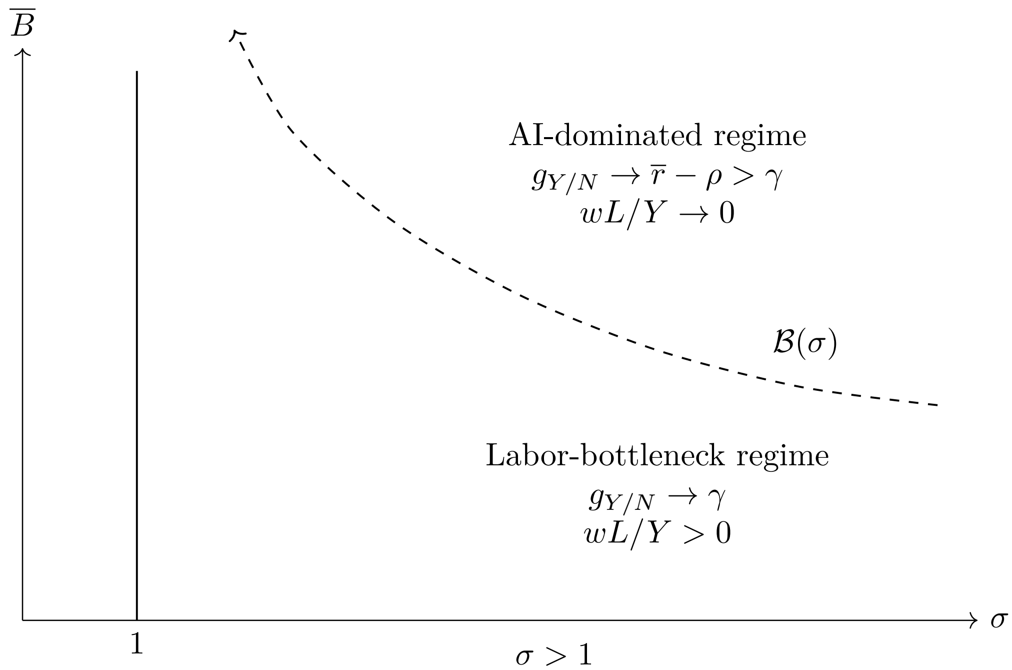

Figure 3: Finite-frontier regimes under gross substitution. The vertical axis is linear and the curve is schematic rather than calibrated. For $\sigma > 1$, $\mathcal{B}(\sigma)$ separates the labor-bottleneck and AI-dominated equilibria constructed in Proposition 3. The dashed boundary itself is not covered by the existence proof.

4.5 Scope of the existence results

Figure 3 summarizes the central comparison for $\sigma > 1$. Here the finite frontier is an analytical device that bounds recursive AI improvement. A frontier below $\mathcal{B}(\sigma)$ supports the labor-bottleneck equilibrium: output per person and the real wage grow at γ, and labor retains a positive income share. Above the threshold, capital and effective AI production services grow faster than effective labor. Output per person then grows faster than γ and labor's income share converges to zero. The figure omits $\sigma < 1$ because I do not interpret $\mathcal{B}(\sigma)$ as a regime boundary under complementarity.

Table 1 collects the sufficient frontier conditions. Every row uses the parameter restrictions in Section 3, together with the initial-stock neighborhood stated in the corresponding proposition. These are not necessary conditions for all possible equilibria, nor a claim of global uniqueness.

Table 1: Constructed finite-frontier equilibria

Elasticity	Frontier condition	$\lim(g_Y - n)$	$\lim wL/Y$
$0 < \sigma < 1$	$\overline{B} > \mathcal{B}(\sigma)$	γ	Positive
$\sigma = 1$	Any finite $\overline{B} > 0$	γ	$(1-\alpha)\omega_L$
$\sigma > 1$	$\overline{B} < \mathcal{B}(\sigma)$	γ	Positive
$\sigma > 1$	$\overline{B} > \mathcal{B}(\sigma)$	$\overline{r} - \rho > \gamma$	Zero

The proof constructs convergent trajectories from nearby predetermined stocks and then verifies both agents' optimality and both transversality conditions. Research remains positive at every finite date. The constants and neighborhoods depend on the parameters; the result does not establish equilibrium from arbitrary K_0, B_0, A_0, N_0, at the threshold, or after taking $\overline{B} \to \infty$. Those are separate questions.

The analytical limits do not require simulations. They do not, however, determine transition duration or the paths of wages, interest rates, and income shares. Claims about a prolonged moderate-growth phase require additional analysis of equilibrium transitions.

4.6 The uncapped economy and the role of the frontier

The comparative static in (50) does not solve the economy with an infinite AI-efficiency frontier. It first fixes a finite $\overline{B}$, constructs an equilibrium, takes that economy's long-run limit, and only then compares the resulting limits as $\overline{B}$ increases. Setting $\overline{B} = \infty$ at the outset changes the state equation by setting $\psi(B) = 1$ and removes the bound on the developer's state. In general,

$$\lim_{\overline{B}\to\infty} \lim_{t\to\infty} r_t(\overline{B}) \quad \text{need not equal} \quad \lim_{t\to\infty} r_t(\infty), \tag{51}$$

and the expression on the right may not exist. The initial-stock neighborhood and the equilibrium choices of C_0 and q_0 also depend on the frontier. Existence for every fixed finite frontier therefore does not imply existence in the uncapped economy.

This distinction is especially important when $\sigma > 1$. At high AI efficiency, effective AI production services can replace labor. Holding current K, A, and L fixed, optimized operating profit then grows asymptotically in proportion to $KB^{(1-\alpha)/\alpha}$. Research raises B, and a higher B both raises operating profit and makes subsequent research compute more productive. The capped equilibria reveal the strength of this feedback: once the economy is in the AI-dominated regime, the limiting interest and output-growth rates increase with $\overline{B}$ and become unbounded as $\overline{B} \to \infty$. This divergence is evidence that the uncapped problem may behave qualitatively differently. It is not, by itself, proof of a finite-time singularity or of equilibrium nonexistence. No analogous divergence appears in the finite-frontier equilibria constructed for $\sigma \leq 1$, where labor remains a long-run bottleneck. This does not establish existence without a frontier in those cases either.

The developer's optimization problem identifies a more immediate possible failure. Proposition 5 in Appendix A shows that, in the uncapped economy, the condition $\eta < \alpha$ makes the benefit generated by a large research expenditure grow at most sublinearly in that expenditure on a fixed finite horizon, while its cost grows linearly. It therefore rules out an arbitrarily large research burst as the developer's optimal response in that partial-equilibrium problem. If instead $\sigma > 1$ and $\eta > \alpha$, the developer can make its finite-horizon payoff arbitrarily large by increasing an early research burst. No finite research plan maximizes its objective. At $\eta = \alpha$, the outcome depends on the coefficients of the leading benefit and cost terms. These results explain why $\eta < \alpha$ is a substantive restriction, but they still do not establish an uncapped infinite-horizon general equilibrium.

The failure when the developer's value is unbounded is economic rather than numerical. Research compute M is a use of the final good, not merely a financial position. The household may finance a negative distribution from the developer, but finance cannot create the real resources used in research. Equation (29a) requires research to compete with consumption, inference compute, and physical investment. If every finite research plan can be improved by choosing a larger research burst, no allocation with finite resources can satisfy the developer's optimality condition and the aggregate resource constraint at the same time. Such an allocation is not an equilibrium under Definition 1.

A different failure can occur even when the developer's value is bounded on every finite horizon. Recursive improvement may cause B, output, capital, or research expenditure to become unbounded at a finite date. I use *finite-time singularity* for this event, rather than for merely rapid growth. A path that ends at such a date is not an equilibrium as defined in Section 3: it does not provide finite, feasible allocations for every $t \geq 0$, and its infinite-horizon optimality and transversality conditions are not defined. The analysis in this paper does not prove that every uncapped trajectory with $\sigma > 1$ reaches such a singularity. It also does not rule out an uncapped equilibrium that remains finite at every finite date. Explosive-growth mechanisms in Nordhaus (2021) and Davidson et al. (2026) motivate this question but do not replace the missing equilibrium argument for the present model.

I therefore use a finite frontier as an analytical continuation device, not as an estimate of a permanent technological ceiling. For each finite $\overline{B}$, the term $1 - B/\overline{B}$ keeps AI efficiency in a bounded state space and makes it possible to establish and compute an equilibrium. I then raise the frontier and examine how the equilibrium regime, growth, interest rates, and income shares change. This strategy separates results that hold for every fixed finite frontier from claims about an uncapped economy. If the capped equilibria fail to converge, or their limiting growth rates diverge, the correct conclusion is that this continuation does not produce an uncapped equilibrium—not that a finite-time singularity has already been proved.

5 Quantitative equilibrium paths

5.1 Parameters and initial conditions

I compare four economies with the same parameters and predetermined initial stocks, varying only σ. Table 2 retains the macroeconomic benchmarks used in the earlier specification. The AI parameters are illustrative, not estimated. All four economies have autonomous research and a positive AI weight; the no-AI economy in Remark 1 is not one of these four scenarios.

Table 2: Parameters and scenario choices

Parameter	Value	Interpretation	Source or rationale
α	0.33	Capital elasticity	Standard one-third macro benchmark
δ	0.05	Annual capital depreciation	Annual benchmark; PWT comparison (Feenstra et al., 2025)
ρ	0.04	Annual household discount rate	Preference calibration
n	0.003	Annual population growth	Constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024)
γ	0.01	Annual growth of labor-augmenting technology	Illustrative long-run trend; see OECD (2025)
ω_X	0.20	Weight on effective AI production services	Scenario choice, not an observed industry share
ω_L	0.80	Weight on effective labor	$1 - \omega_X$
η	0.20	Research exponent	Illustrative choice; not estimated
χ	1.4378	Research productivity	Calibrated transition-speed parameter
σ	0.90; 1; 1.10; 1.50	AI–labor substitution	Four scenarios spanning the analytical regimes
$\overline{B}$	$1.10\,\mathcal{B}(1.50)$ $\simeq 170.124$	Common AI-efficiency frontier	Ten percent above the threshold for $\sigma = 1.50$

The constant n extrapolates a finite-period demographic approximation; it is not a claim that the UN projects constant population growth forever. The OECD benchmark for γ concerns measured labor-productivity growth, not A directly. I use it only as an approximate guide to the magnitude of long-run labor-augmenting technological progress. The parameter χ determines research speed relative to the other annual rates, so it is not merely a free time-unit normalization. I choose $\chi = 1.4378$ so that, in the $\sigma = 1.50$ equilibrium, the labor income share completes half of the distance between its date-zero value and its analytical limit after 50 model years. This is an illustrative timing normalization, not an estimate. Different values of χ produce faster or slower transitions and require the entire equilibrium path to be solved again. The horizontal axes below measure model years, not calendar forecasts. Likewise, the OECD's discussion of AI-investment measurement supports treating ω_X as a scenario parameter, but does not identify its value (Fonteneau et al., 2025; OECD, 2026).

The common frontier is below $\mathcal{B}(1.10) \simeq 60.4$ million but above $\mathcal{B}(1.50) \simeq 154.658$. Thus $\sigma = 1.10$ retains a positive long-run labor share, whereas $\sigma = 1.50$ enters the AI-dominated regime. Changing the frontier separately across scenarios would confound this comparison of elasticities.

The choice of initial conditions follows the question posed by each exercise. $A_0 = N_0 = 1$ fixes units. The capital stock K_0 and AI efficiency B_0 are predetermined states and are therefore specified before solving an equilibrium. Consumption C_0 and the developer's shadow value q_0 are jump variables: I never import them from another economy or choose them to improve numerical convergence. The BVP selects them anew for every reported path.

I set

$$A_0 = N_0 = 1, \qquad K_0 = 4, \qquad B_0 = 0.10\overline{B} \simeq 17.0124. \tag{52}$$

These common initial stocks place the economy closer to the capped $\sigma = 1$ terminal regime than the earlier uncapped reference: initial AI efficiency is 10 percent of its frontier and initial capital per unit of effective labor is about 44 percent of its terminal $\sigma = 1$ value. I use these intermediate stocks so that the main figures display the transition without letting the financing needs of an extremely distant initial economy dominate the comparison. Holding the same two stocks across the four values of σ then isolates how the substitution elasticity changes the subsequent equilibrium path. These values are illustrative rather than estimated, and they are not a balanced-growth state of the capped economy. An exact constant-efficiency reference at $B_0 = \overline{B}$ would eliminate further improvements in AI efficiency and would lie outside the interior problem studied here. In each capped economy, the BVP selects C_0 and q_0 anew; neither jump variable is fixed at its uncapped reference value.

5.2 Numerical method and equilibrium admission

I retain the four-dimensional boundary-value architecture: two initial stock conditions and two terminal conditions selecting the stable equilibrium continuation. The appropriate finite-frontier regime from Section 4, not the uncapped unit-elastic BGP, determines the terminal conditions. Appendix B documents the continuation, equation reconstruction, and numerical checks.

In economic terms, the algorithm takes K_0 and B_0 as given and selects the initial consumption and shadow value that place the economy on the stable equilibrium path. It first constructs that path near the analytically characterized terminal regime and then moves it gradually to the prescribed initial stocks. Solving the boundary-value problem is only the construction step; the separate admission tests determine whether the resulting candidate is reported as an equilibrium.

A converged BVP is not sufficient for admission. I also check feasibility, independently reconstructed dynamic residuals, stability to longer horizons and tighter tolerances, both transversality conditions, and sufficient global developer optimality. The first three scenarios satisfy the global concavity test. For $\sigma = 1.50$, that stronger test fails at low counterfactual AI-efficiency levels, but the support condition in Appendix Lemma 2 is satisfied. Its analytical long-run bound is given in (80).

5.3 Growth, prices, and the distribution of output

Figures 4–6 present the transition in three steps: quantities and accumulation, prices and returns, and the distribution of output. Each panel includes the same four admitted trajectories. The three labor-bottleneck paths consequently appear close to one another in several panels. This compression reflects their common long-run regime; exact initial and limiting values of the central outcomes are reported in Table 3.

The relative growth of effective AI production services is central to reading the figures. Rearranging (10) gives

$$\frac{s_X}{1 - s_X} = \frac{\omega_X}{\omega_L} \left(\frac{X}{AL} \right)^{(\sigma - 1)/\sigma} . \tag{53}$$

Differentiating this expression yields

$$\frac{d}{dt} \log\left(\frac{s_X}{1 - s_X} \right) = \frac{\sigma - 1}{\sigma} \left[g_X - (n + \gamma) \right] . \tag{54}$$

Thus the growth of $X/(AL)$ determines the direction and speed at which the AI share changes, conditional on σ.

Figure 4 first reports the growth rates of output, effective AI production services, and capital per unit of effective labor. These are $g_Y - (n + \gamma)$, $g_X - (n + \gamma)$, and $g_K - (n + \gamma)$, respectively. Growth rates show when the economy accelerates without compressing the large level differences that emerge across regimes.

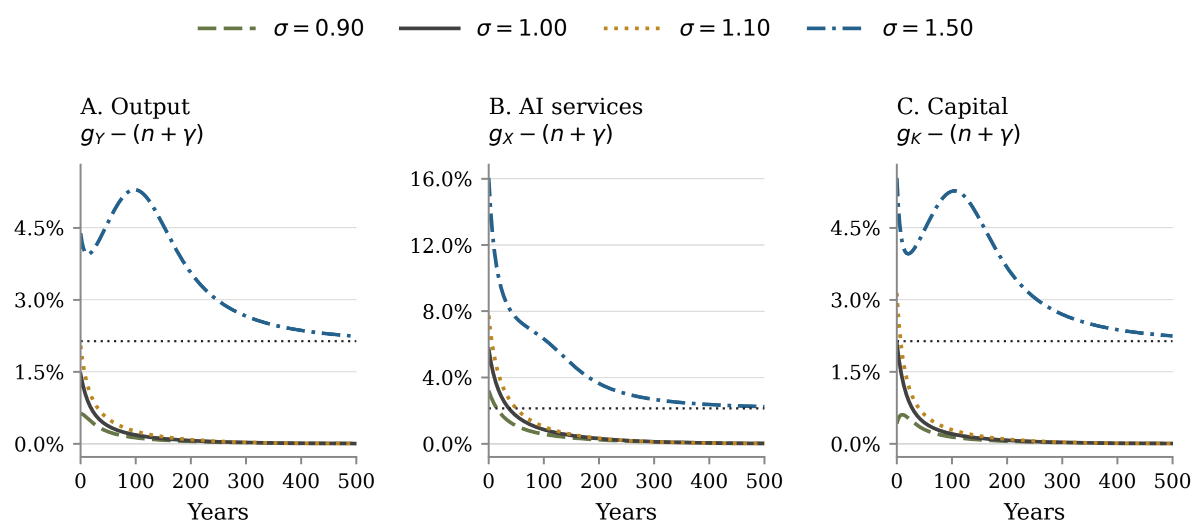

Figure 4: Growth of output, effective AI production services, and capital per unit of effective labor. Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Figure 5 next reports real-wage growth, the net interest rate, and the price of effective AI production services. The first two are rates; p_X is a relative-price level measured in units of the final good. Condition (19) implies $p_X = 1/[B(1-e_X)]$. Because B approaches its finite frontier and e_X approaches a value below one in every terminal regime, p_X converges to a positive finite level. It falls rather than grows without bound in all four simulations. I therefore report its level on a logarithmic scale, which keeps the four positive price paths visible despite their different magnitudes.

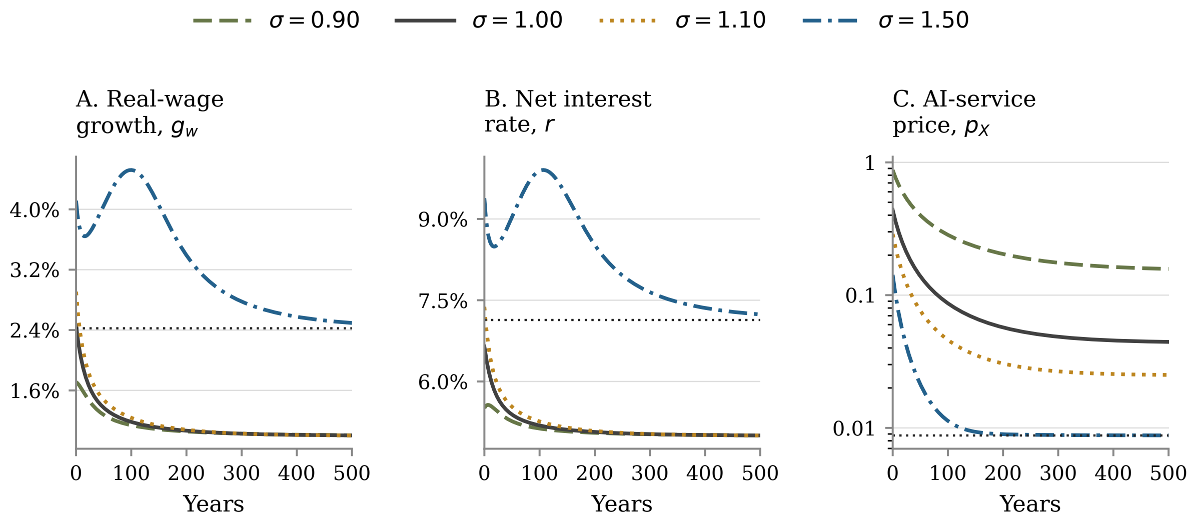

Figure 5: Wages, returns, and the AI-service price. Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years. Real-wage growth and the net interest rate use linear percentage scales. The level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Finally, Figure 6 reports labor income and the three uses of AI-industry revenue as shares of final output. Combining final-good zero profit with $\Pi = p_X X - U - M$ gives

$$\frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y} = 1 - \alpha. \tag{55}$$

The four panels therefore exhaust the noncapital share of output. Gross capital income, $(r + \delta)K/Y = \alpha$, is constant and accounts for the remainder.

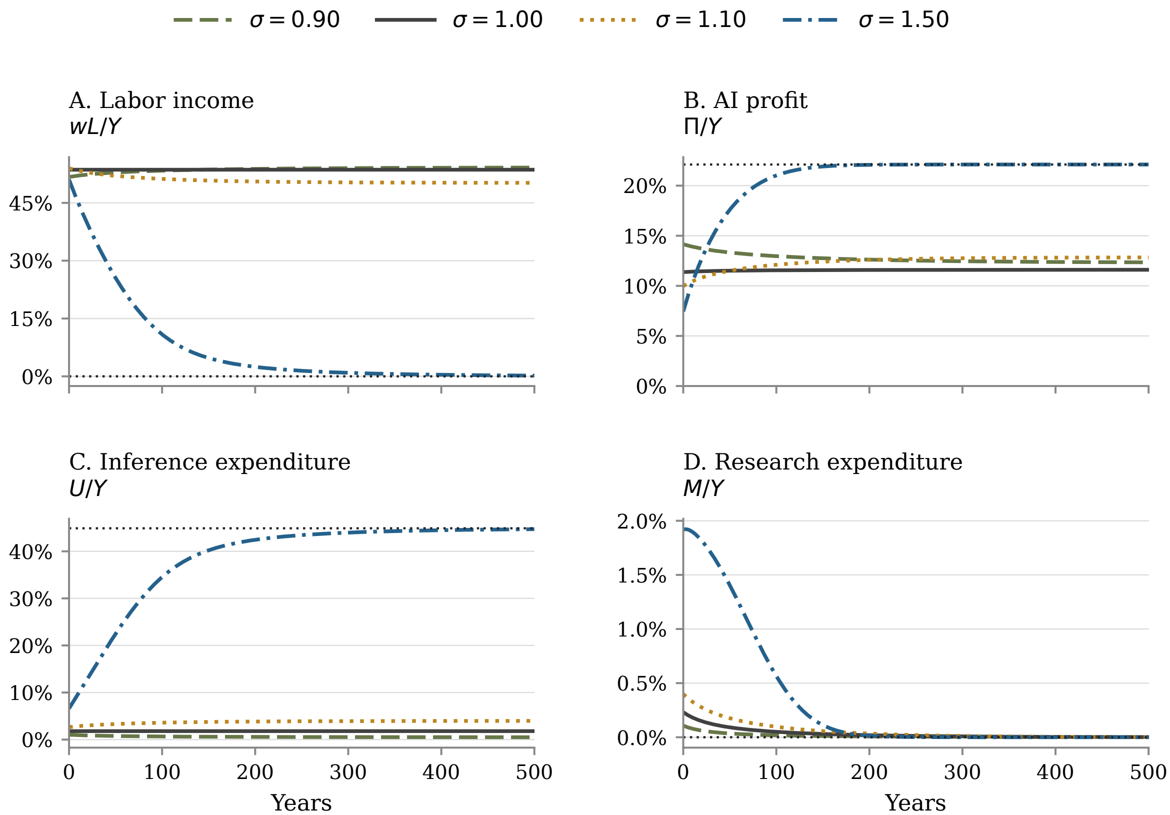

Figure 6: Distribution of the noncapital share of output. The panels report labor income, AI profit, inference expenditure, and research expenditure, each divided by final output. Together they sum to $1-\alpha$; the omitted gross capital-income share is the constant α. Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years and uses a linear percentage scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Table 3: Initial and long-run values of the central outcomes

σ	Date	$g_Y - n$	g_w	r	wL/Y	$p_X X/Y$
0.90	0	1.629	1.711	5.508	51.7	15.3
	∞	1.000	1.000	5.000	54.2	12.8
1.00	0	2.458	2.458	6.646	53.6	13.4
	∞	1.000	1.000	5.000	53.6	13.4
1.10	0	3.046	2.909	7.380	54.0	13.0
	∞	1.000	1.000	5.000	50.2	16.8
1.50	0	5.390	4.115	9.385	51.0	16.0
	∞	3.135	2.423	7.135	0.0	67.0

Note: All entries are percentages. Date-zero entries come from the numerical equilibrium paths. Infinite-date entries are the analytical limits characterized in Section 4.

The first three scenarios approach per-person growth of 1% and a net interest rate of 5%. For $\sigma = 1.50$, their analytical limits are 3.135% and 7.135%, respectively. Real wages then grow asymptotically at 2.423%. Thus output per person grows 0.712 percentage points faster than the real wage. By Proposition 4, this implies an asymptotic decline of 0.712% per year in the log labor income share. The central distributional result is therefore not that real wages decline: workers become richer in absolute terms while their claim on total output vanishes. AI revenue meanwhile approaches 67% of output. These limiting rates describe the equilibrium continuations; they are not imposed as constant rates during the transition.

The AI-dominated path also displays signals before labor's income share has fallen sharply. At date zero, $X/(AL)$ grows at 16.051% and the net interest rate is 9.385%, well above the 5% labor-bottleneck benchmark, while labor still receives 51.0% of output. Moreover, because $L = N$, the accounting identity

$$\frac{d}{dt}\log\left(\frac{wL}{Y}\right) = g_w - (g_Y - n) \tag{56}$$

holds at every date. The gap between output-per-person and real-wage growth therefore does not merely predict a later distributional change: a positive gap is exactly the contemporaneous rate at which the log labor income share declines, and a widening gap means that decline is accelerating.

Equations (53) and (54) provide the structural explanation for the speed of the transition. A value $\sigma > 1$ makes the AI share rise with $X/(AL)$, but it does not make AI immediately replace labor. When σ is close to one, the exponent $(\sigma - 1)/\sigma$ is small and the share responds slowly even to large increases in relative AI services. At $\sigma = 1.50$, for example, the exponent is $1/3$: increasing the odds $s_X/(1-s_X)$ by a factor of ten requires a thousand-fold increase in $X/(AL)$. For elasticities closer to one, the required increase is larger still, conditional on the frontier being high enough to support the AI-dominated regime.

The economy must also accumulate the inputs that produce this change. Neither K nor B can jump. Expanding X requires inference compute $U = X/B$, while improving B requires research compute M; both uses compete with consumption and physical investment for current output. Moreover, $\eta = 0.20$ implies strong diminishing returns to a contemporaneous increase in research compute. As B rises, however, it reduces the compute required to supply a unit of X and makes subsequent research compute more productive. The feedback therefore strengthens gradually. Near the finite frontier, $\psi(B)$ slows further efficiency improvement. The eventual acceleration in the AI-dominated equilibrium is consequently not an explosion of B: it is sustained by the expansion of capital and effective AI production services after labor ceases to be the binding input.

The date of takeoff is not a structural prediction of the model. Equation (53) implies an earlier transition when σ is farther above one or when X grows faster relative to AL. Within the model, greater research productivity χ, a higher initial B_0, or a larger AI weight ω_X can advance the transition through different channels. Faster capital accumulation or slower growth of effective labor can also allow $X/(AL)$ to rise sooner. These changes are not interchangeable: σ and ω_X alter the production technology, B_0 changes the initial economy, and χ changes the research clock. The effects of changing η or $\overline{B}$ on the takeoff date do not have an unambiguous general-equilibrium sign because they also change research incentives and the terminal equilibrium. These comparative mechanisms indicate how the transition may move, but do not identify a date without recalculating the equilibrium path.

The $\sigma = 1.50$ path displays this mechanism under the calibrated research speed. The labor share completes 10 percent of its decline after 8.3 model years, half after 50 years, and 90 percent after 145.2 years. At year 50, output per person grows at 5.616%, the real wage at 4.050%, and labor receives 25.5% of output. At year 100, the corresponding growth rates are 6.290% and 4.522%, while labor's share has fallen to 10.9%. By year 500, output-per-person growth is 3.239%, real-wage growth is 2.494%, and labor's share is 0.19%; all three are close to their analytical limits. The non-monotone growth response reflects the costly initial research surge, the subsequent acceleration in effective AI services, and eventual convergence as B approaches its frontier. These dates illustrate an equilibrium under one research-speed calibration. They are not estimates of when an actual AI transition will occur.

Substitution alone does not produce this outcome. With the same frontier and $\sigma = 1.10$, the labor share instead approaches 50.2%, compared with 53.6% at $\sigma = 1$ and 54.2% at $\sigma = 0.90$. The frontier's position relative to the elasticity-specific threshold is essential to interpreting the comparison.

5.4 A transition from the no-AI steady state

As an additional transition experiment, suppose that the economy is initially on the no-AI Ramsey–Cass–Koopmans steady state described in Remark 1. Before date zero, $\omega_X = 0$ and $\omega_L = 1$. AI efficiency is then irrelevant for production, so I assign it a low latent value, $B_0/\overline{B} = 0.01$, solely to initialize the economy after AI production and research become available. The no-AI steady-state capital stock and the two predetermined date-zero stocks are

$$\begin{aligned} \frac{K}{AN}\bigg|_{\mathrm{RCK}} &= \left(\frac{\alpha}{\rho+\delta+\gamma}\right)^{1/(1-\alpha)} = 5.9416, \\ A_0 = N_0 = 1, &\qquad K_0 = 5.9416, \quad B_0 = 0.01\overline{B} = 1.7012. \end{aligned} \tag{57}$$

The corresponding pre-change values are $Y/(AN) = 1.8005$ and $C/(AN) = 1.4262$. At date zero, ω_X rises to 0.20 and ω_L falls to 0.80. I hold the two stocks in (57) fixed across the four values of σ. The value of K_0 is therefore not an additional calibration: it is the exact no-AI steady-state capital stock implied by the maintained macro parameters. The one-percent value for $B_0/\overline{B}$ is an illustrative low starting point, chosen to leave substantial room for AI improvement after the technology switch; it cannot be inferred from the pre-change economy because B has no economic role when $\omega_X = 0$. Keeping these two predetermined stocks common across σ isolates the role of substitution conditional on the same Ramsey capital stock and the same latent initial AI efficiency. Consumption can jump after the change in technology, and the developer's shadow value has no no-AI counterpart. The BVP therefore selects C_0 and q_0 anew in each positive-AI economy.

Figures 7– 9 use the same variables, normalizations, scenario order, and scale conventions as the main comparison. All four paths are independently solved and pass the equilibrium-admission tests.

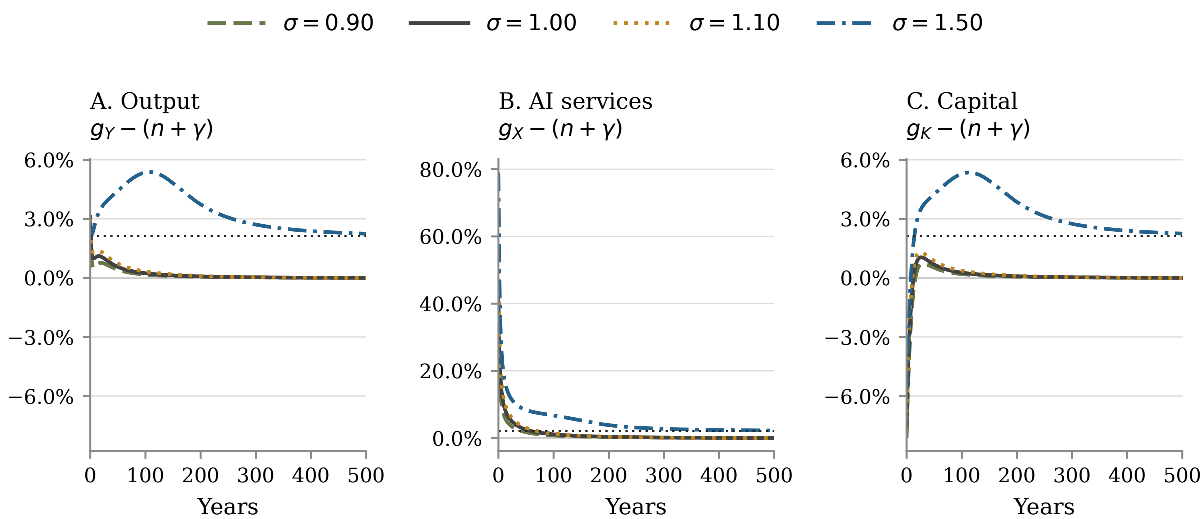

Figure 7: Growth of output, effective AI production services, and capital per unit of effective labor after the switch from the no-AI steady-state stocks. Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

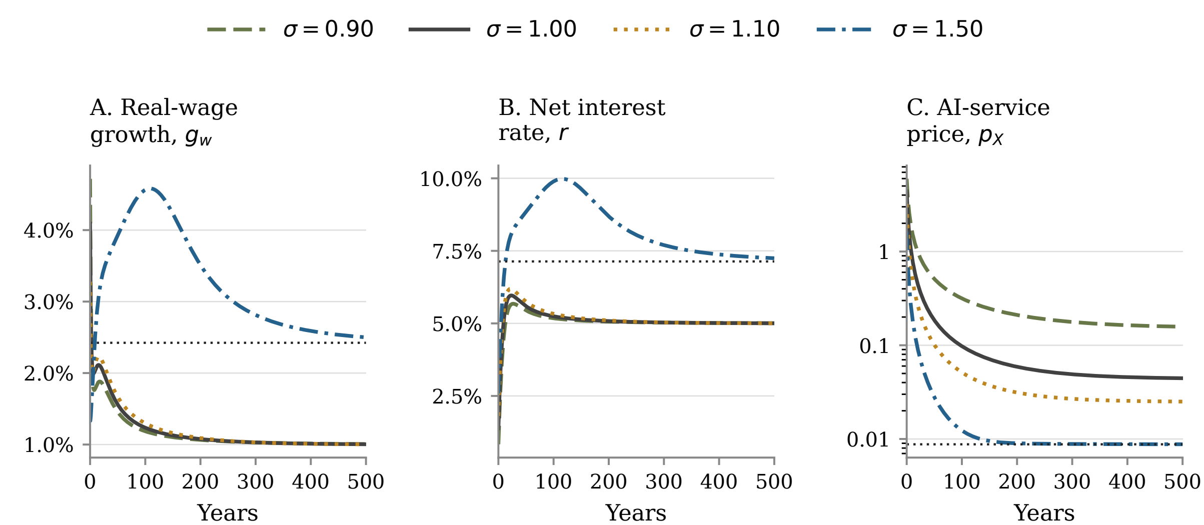

Figure 8: Wages, returns, and the AI-service price after the switch from the no-AI steady-state stocks. Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Real-wage growth and the net interest rate use linear percentage scales; the level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

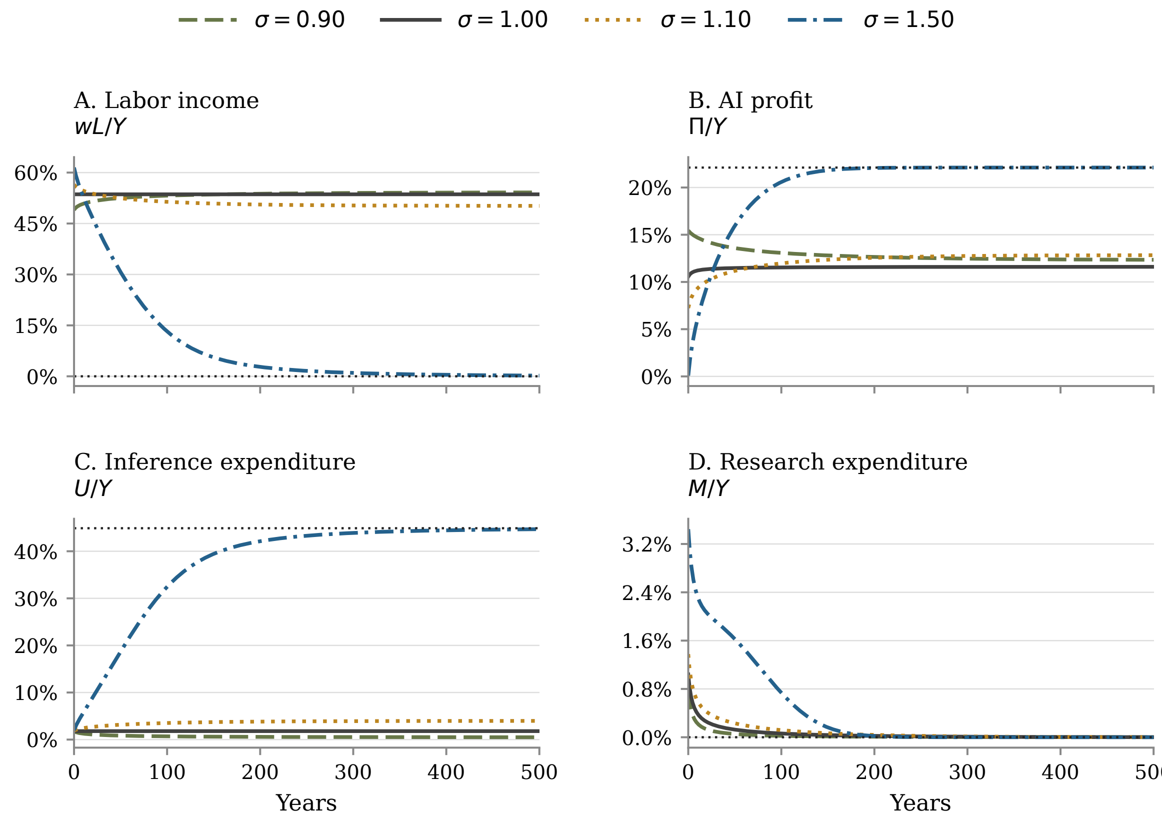

Figure 9: Distribution of the noncapital share of output after the switch from the no-AI steady-state stocks. Labor income, AI profit, inference expenditure, and research expenditure are divided by final output and sum to $1-\alpha$. The omitted gross capital-income share is the constant α. Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Low initial AI efficiency produces a sharp reallocation. At date zero, $X/(AL)$ grows between 30.5% and 79.0% across the four equilibria, while $K/(AL)$ falls between 5.7% and 8.1%. The net interest rate is initially between 0.83% and 2.59%, below its 5% no-AI value. These movements do not imply that access to an additional technology must lower current output. The experiment changes the CES weights from $(\omega_L, \omega_X) = (1, 0)$ to $(0.80, 0.20)$, so the old production technology does not remain available as an outside option. It should therefore be read as a change in technological regime from Ramsey initial stocks, not as a welfare comparison that holds the pre-change production set fixed.

The high-substitution economy subsequently separates from the other three. For $\sigma = 1.50$, labor initially receives 61.5% of output. Its share completes half of the decline toward zero after approximately 50 model years and falls to 0.2% by year 500. Output-per-person growth peaks at 6.38% around year 107, real-wage growth peaks at 4.58% around year 109, and the net interest rate peaks at 9.98% around year 116. By year 500, the three rates are 3.25%, 2.50%, and 7.25%, respectively, and continue toward their analytical limits. In the other three economies, output-per-

person and real-wage growth have returned to approximately 1% and the net interest rate to 5% by year 500. The same Ramsey initial capital stock can therefore lead either back to a labor-bottleneck regime or toward the AI-dominated regime; the difference is the elasticity and the frontier relative to its elasticity-specific threshold.

5.5 Near-terminal paths

The initially high and declining growth rates of the three labor-bottleneck paths in the main comparison are not their long-run prediction. They partly reflect the distance between the common initial stocks in (52) and each economy's terminal allocation. To isolate terminal behavior while preserving the comparison across regimes, I solve an additional exercise for all four values of σ. I set

$$\frac{B_0}{\overline{B}} = 0.9999, \qquad \frac{K_0}{A_0 N_0} = \frac{K}{AN}\bigg|_{\text{terminal}} = \begin{cases} 6.5571, & \sigma = 0.90, \\ 9.0990, & \sigma = 1, \\ 12.2652, & \sigma = 1.10. \end{cases} \tag{58}$$

for the three labor-bottleneck paths. In the AI-dominated regime, $K/(AN)$ has no finite terminal value. I therefore initialize the predetermined state that does converge in Proposition 3:

$$(\overline{B} - B_0)K_0 = \overline{d}, \qquad K_0 = 4{,}101{,}704.23 \quad \text{when } \sigma = 1.50. \tag{59}$$

Consumption and the developer's shadow value are again selected by the BVP for each path. I use $B_0/\overline{B} = 0.9999$, rather than the frontier itself, to begin sufficiently close to the analytical limits while preserving the interior research problem. For each labor-bottleneck path, setting capital at its terminal normalized value removes a separate capital-adjustment gap. For $\sigma = 1.50$, the corresponding stationary object is $(\overline{B} - B)K$, not $K/(AN)$, which explains the different capital rule in (59). These are diagnostic initial conditions chosen to display terminal behavior; they are not empirical calibrations. Because $B_0 < \overline{B}$ and research remains positive, these are not exact steady states at a finite date. They are equilibrium paths that begin near their respective terminal regimes. The initial capital condition varies across regimes because their convergent state variables differ. This exercise therefore isolates terminal behavior; it does not hold all initial stocks fixed.

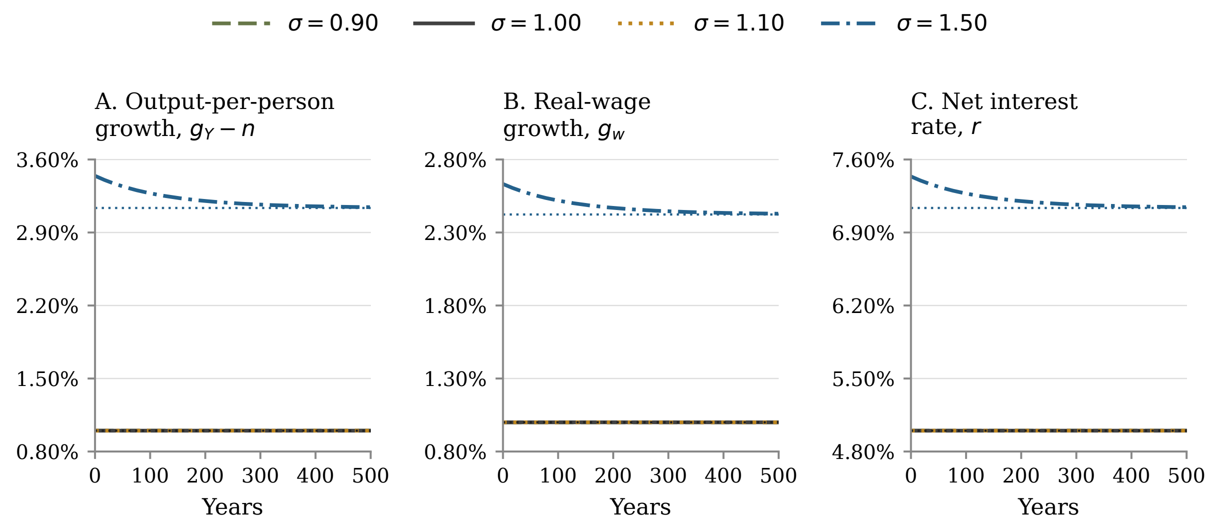

Figure 10: Near-terminal equilibrium paths. The panels report output-per-person growth, real-wage growth, and the net interest rate for four independently admitted numerical equilibria over 500 model years. Dark horizontal dotted lines mark the common labor-bottleneck limits; blue dotted lines mark the analytical limits for $\sigma = 1.50$.

Figure 10 removes the large date-zero movements of the three labor-bottleneck paths in the main comparison. Their output-per-person growth, real-wage growth, and net interest rate all begin within 0.001 percentage points of their analytical limits of 1%, 1%, and 5%. For $\sigma = 1.50$, the corresponding date-zero values are 3.443%, 2.632%, and 7.437%; by year 500 they have reached 3.143%, 2.429%, and 7.143%, close to their analytical limits of 3.135%, 2.423%, and 7.135%. The exercise confirms that the high initial growth rates in the main labor-bottleneck paths are transitional responses to the common starting stocks. It also displays the distinct AI-dominated long run without the much larger initial adjustment present in the main comparison.

5.6 A slower transition

The 50-year midpoint in the main comparison is a timing calibration rather than a prediction. To show how strongly the transition can stretch out, I recover the earlier calibration with

$$\chi = 0.01, \qquad K_0 = 2.027733654, \qquad B_0 = 0.443670932. \tag{60}$$

All other parameters, the common frontier, and the four values of σ are unchanged. These initial stocks are the values implied by the earlier uncapped unit-elastic BGP. Together with the low value $\chi = 0.01$, they place the capped economy far from its frontier and are chosen to ask whether the same long-run regimes can remain hidden for much longer. They are a sensitivity design, not estimates of the current world economy. Neither the initial consumption nor the initial shadow value is imported from the uncapped BGP. Both jump variables are solved again for each capped economy. The exercise changes both research productivity and the initial distance from the frontier, so it should not be interpreted as a one-parameter estimate of the effect of χ. Each of the four paths is independently re-solved and passes the same equilibrium-admission tests as the main comparison.

Figures 11– 13 retain the variables, normalizations, scenario order, and scale conventions of the main figures. The only presentational change is the 4,000-year window needed to display the slower adjustment.

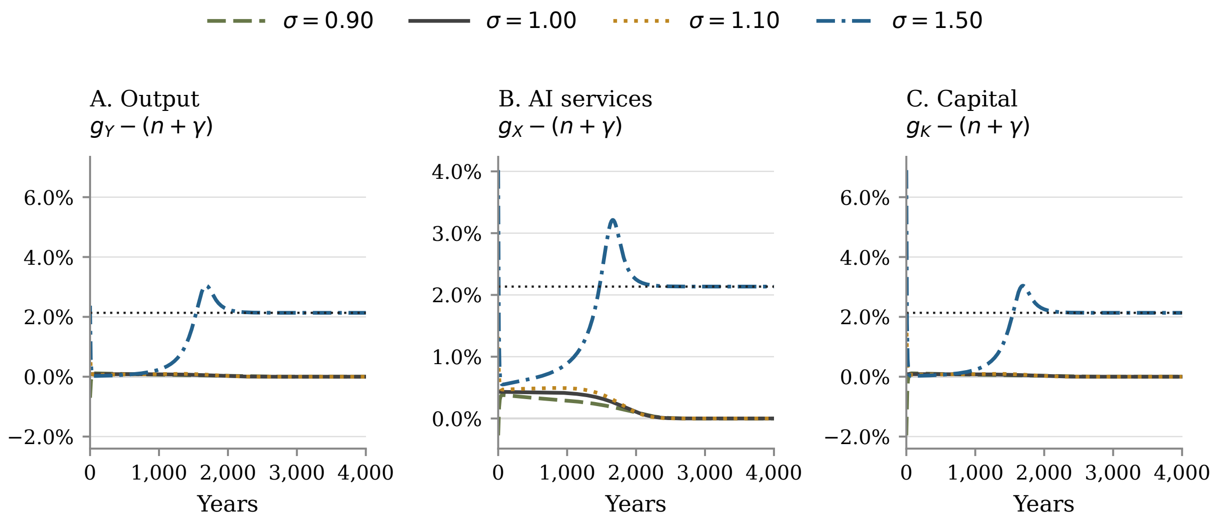

Figure 11: Growth of output, effective AI production services, and capital per unit of effective labor under the slower-transition calibration. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

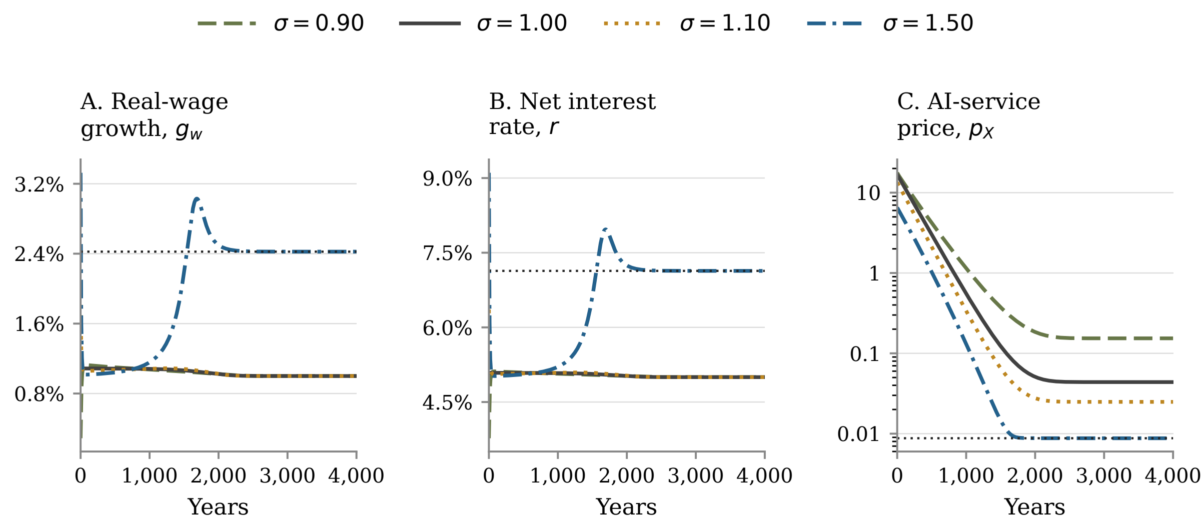

Figure 12: Wages, returns, and the AI-service price under the slower-transition calibration. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Real-wage growth and the net interest rate use linear percentage scales; the level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

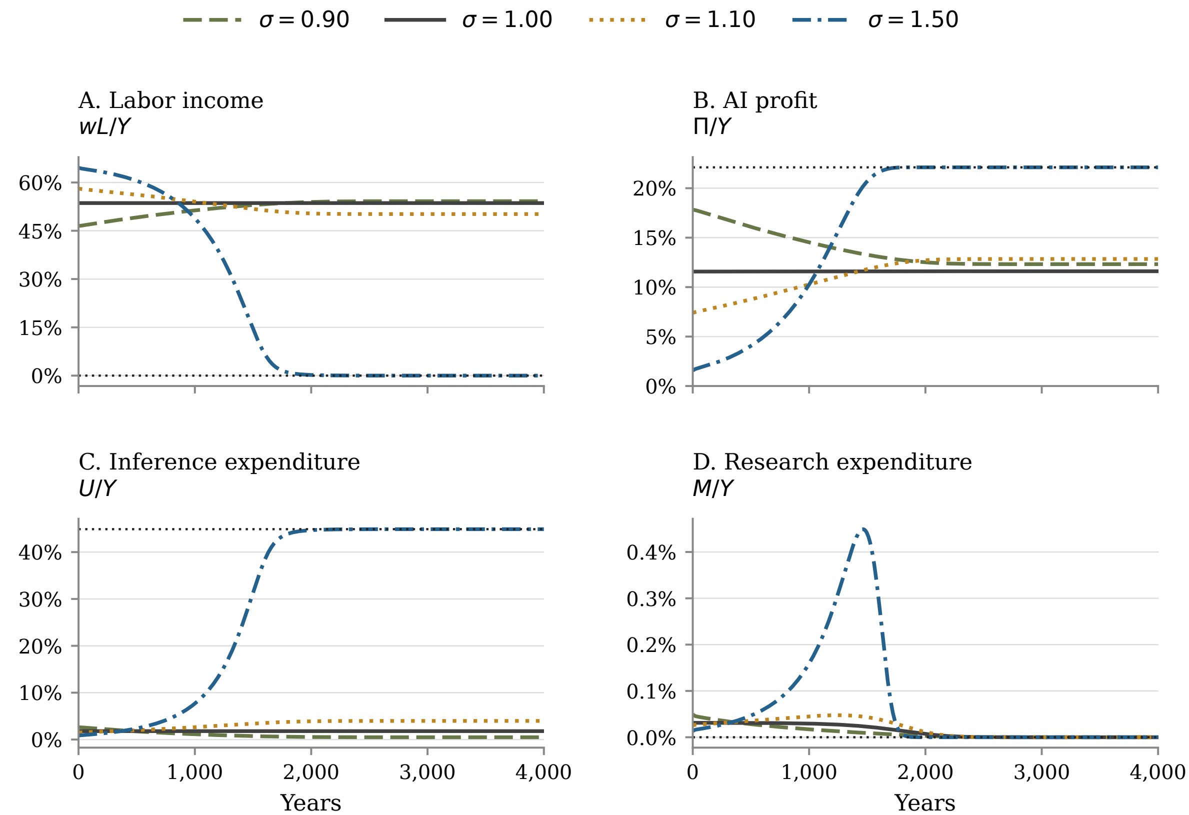

Figure 13: Distribution of the noncapital share of output under the slower-transition calibration. Labor income, AI profit, inference expenditure, and research expenditure are divided by final output and sum to $1 - \alpha$. The omitted gross capital-income share is the constant α. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

In the slow $\sigma = 1.50$ equilibrium, labor's income share completes only 10 percent of its decline after 657 model years, half after 1,295 years, and 90 percent after 1,605 years. At year 500, output per person grows at 1.062%, the real wage at 1.043%, and labor still receives 60.5% of output. By year 1,500, output-per-person growth has risen to 2.797%, while labor's share has fallen to 14.5%. The analytical limits are the same as in the main comparison because σ and $\overline{B}$ are unchanged; what changes radically is when the economy approaches them. The model can therefore generate a long interval in which little appears to happen before growth and the distribution of output change sharply. It does not identify which of the two timing calibrations is empirically more plausible.

5.7 Initial financing sensitivity

Moving the common initial stocks closer to the $\sigma = 1$ terminal regime materially reduces the initial financing requirement. In the reported $\sigma = 1.50$ path, research absorbs 12.0% of AI revenue, inference absorbs 41.4%, and the developer distributes the remaining 46.6% as profit. These quantities equal 1.92%, 6.60%, and 7.44% of final output, respectively.

This benign date-zero composition is not invariant to the initial stocks. In an admitted sensitivity path, I deliberately reuse the more distant stocks $K_0 = 2.0277$ and $B_0/\overline{B} = 0.26\%$ from the earlier numerical design. This choice isolates the financing consequences of beginning farther from the terminal regime. I then recalibrate χ to preserve the same 50-year transition midpoint as in the main $\sigma = 1.50$ path, so the comparison does not merely contrast different transition dates. The sensitivity path is a separately solved equilibrium, and its C_0 and q_0 are again selected by the BVP. Research expenditure initially equals 106.0% of AI revenue and profit is -40.5% of AI revenue. The household owner then finances the loss through an equity injection because current research raises B and the present value of future monopoly rents. The benchmark permits this intertemporal financing: it imposes neither a borrowing constraint nor a nonnegative-profit constraint on the developer. Adding such a constraint could delay or restrict the transition. The contrast therefore identifies initial technological distance and financing capacity as substantive determinants of the transition, not as innocuous plotting choices.

In the AI-dominated limit, inference absorbs 67% of AI revenue, research's revenue share vanishes, and the developer retains 33%. Its net distribution therefore approaches 22.11% of final output. By contrast, at $\sigma = 1$ AI revenue is always 13.4% of output and inference absorbs 13.4% of that revenue. A vanishing research share does not mean that research stops: the constructed finite-frontier equilibria have a positive limiting level of research compute, while industry revenue continues to grow.

6 Conclusion

The endogenous growth model showed that the long-run economic role of human labor depends critically on whether AI can replace labor's contribution to production. It characterizes how this substitution margin shapes economic growth, wages, interest rates, and the functional distribution of income when AI research is autonomous.

When AI and labor are complements, human labor remains a bottleneck. Improvements in AI efficiency raise productive capacity but do not permanently increase the growth of output per person or real wages beyond the exogenous rate of labor-augmenting technological progress. Labor therefore retains a positive share of income. The same broad conclusion holds under unit elasticity: AI can support sustained growth, but human labor remains economically essential and continues to receive a positive share of output.

The outcome changes when AI and labor are sufficiently substitutable. Gross substitution is necessary but not sufficient: labor remains a bottleneck while AI services are not productive enough to replace its contribution. Once AI becomes sufficiently efficient, however, capital and effective AI production services can expand faster than effective labor. Output per person and real wages then grow faster than in the labor-bottleneck regime, and the interest rate is higher.

Across the long-run regimes I characterize, real wages grow faster than the exogenous rate only when labor's income share vanishes. The declining share does not generate faster wage growth.

In the AI-dominated regime, output per person grows rapidly enough to raise real wages despite the shrinking fraction paid to labor. Because output per person grows even faster than the real wage, labor's income share converges to zero. The gross capital share is fixed by the production technology, so this transformation reallocates the remainder of output from labor toward the AI industry. AI-industry revenue should not be confused with monopoly profit: it finances inference and research before the residual is distributed to the developer's owner.

The transition can be as important as the long-run outcome. In the calibrated AI-dominated path, labor completes half of its income-share decline after 50 model years while output and wages grow rapidly. This timing is illustrative: research productivity is not estimated, and changing it produces faster or slower equilibrium adjustments rather than merely relabeling time. The initial technological distance matters as well. The simulations nevertheless reveal signals before most of the decline in labor's income share occurs: effective AI production services can grow rapidly, the interest rate can rise above the labor-bottleneck benchmark, and a positive output–wage growth gap shows that labor's income share is already falling. Starting farther from the terminal regime can require the developer's owner to finance large temporary losses against future monopoly rents.

The analysis points to three lines of future work. A task-based extension with heterogeneous workers, new human tasks, and a continuing role for human researchers could combine the displacement and reinstatement mechanisms of Acemoglu and Restrepo (2019), the labor-market transitions studied by Korinek et al. (2026), and the research-automation network of Davidson et al. (2026). An explicit upstream compute sector, alternative market structures, and heterogeneous asset ownership could build on Korinek and McKelvey (2026) and Korinek and Vipra (2025) and translate changes in functional income shares into household inequality and economic power. Finally, a more complete existence analysis could remove the finite AI-efficiency frontier, or make its level endogenous, and determine whether the resulting economy admits an equilibrium. This question is related to, but not resolved by, the explosive-growth mechanisms in Nordhaus (2021) and Davidson et al. (2026).

The central question is whether human labor remains sufficiently essential for wages to keep pace with the economy as a whole.

A Proofs

Finite-frontier invariance. Let $D = \overline{B} - B$. The finite-frontier state equation gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{\overline{B}}(BM)^{\eta}.$$

Integration gives

$$\overline{B} - B_t = (\overline{B} - B_0)\exp\left\{-\frac{\chi}{\overline{B}}\int_0^t (B_s M_s)^{\eta}\,\mathrm{d}s\right\} > 0. \tag{61}$$

Local integrability makes the exponent finite at every finite date, so $D_t > 0$.

Lemma 1 (Static AI-service choice). *For $K, A, L, B > 0$, $0 < \alpha < 1$, $\sigma > 0$, and strictly positive CES weights that sum to one, problem (18) has a unique, strictly positive global maximizer. It is characterized by condition (19) and depends smoothly on the positive variables K, A, L, B, for each fixed σ.*

Proof of Lemma 1. Fix $K, A, L, B > 0$, let $R(X) = p_X(X)X$, and write

$$u = \frac{1}{\sigma}, \qquad s = s_X, \qquad e = u(1-s) + \alpha s.$$

The demand elasticity in (15) and the CES-share derivative imply

$$R'(X) = p_X(1-e), \qquad \frac{\mathrm{d}s}{\mathrm{d}\ln X} = (1-u)s(1-s).$$

Let $m(X) = R'(X)$. Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{Xm'(X)}{p_X} = e(1-e) + (\alpha - u)(1-u)s(1-s) \equiv D(s). \tag{62}$$

If $(\alpha - u)(1-u) \geq 0$, then $D(s) > 0$ immediately. If this product is negative, the parameter restrictions imply $0 < \alpha < u < 1$, and

$$D(s) = u(1-u)(1-s)^2 + [u^2 + 2\alpha - 3\alpha u]s(1-s) + \alpha(1-\alpha)s^2.$$

The middle coefficient is positive. For $u \leq 2/3$, this follows from $u^2 + \alpha(2-3u) > 0$. For $u > 2/3$, use $\alpha < u$ and $2 - 3u < 0$ to obtain

$$u^2 + \alpha(2-3u) > u^2 + u(2-3u) = 2u(1-u) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive. At a solution of $m(X) = 1/B$, its derivative is strictly negative. The implicit-function theorem therefore makes the maximizing service choice smooth in K, A, L, B wherever it exists.

For $\sigma = 1$, define $\beta = (1-\alpha)\omega_X$. As $X \downarrow 0$, revenue is proportional to X^{1-u} when $\sigma > 1$, to X^β when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $m(X) \to \infty$. As $X \to \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $m(X) = 1/B$ has exactly one solution. Operating profit increases before this solution and decreases after it. The linear cost X/B dominates revenue as $X \to \infty$, so this solution is the unique global maximum. Equation (62) also proves the strict second-order condition:

$$e_X(1-e_X) + (\alpha - \sigma^{-1})(1-\sigma^{-1})s_X(1-s_X) > 0. \tag{63}$$

□

Interior research. Lemma 1 establishes $X^* > 0$, and the envelope condition (20) gives $\pi_B = X^*/B^2 > 0$. A marginal increase in AI efficiency therefore raises current optimized profit and weakly raises future AI efficiency under a fixed research policy. At a differentiable optimum its current-value shadow price is strictly positive: $q > 0$. The part of the Hamiltonian that depends on M is

$$-M + q\chi B^\eta M^\eta \psi(B).$$

Its right derivative diverges as $M \downarrow 0$ because $q, B, \psi(B) > 0$ and $0 < \eta < 1$. Therefore $M = 0$ cannot maximize the Hamiltonian at a finite date.

Research expenditure and finite-horizon value. The first-order and transversality conditions do not by themselves establish that the developer's objective has a finite value or that a candidate policy is globally optimal. The result below addresses the value on a fixed finite horizon, taking the paths of K, A, L, and r as given. I retain $0 < \eta < 1$ from the AI-efficiency law and distinguish the uncapped cases $\eta < \alpha$, $\eta = \alpha$, and $\eta > \alpha$. Global optimality is addressed separately in Lemma 2.

Proposition 5 (Large-scale research expenditure)**.** *Fix a finite horizon, an initial AI-efficiency level, positive continuous paths for K, A, L, and an integrable path for r. In the uncapped economy with $\overline{B} = \infty$, if $0 < \eta < \alpha < 1$, the developer's truncated value is finite and its objective tends to minus infinity as total research expenditure diverges. If $\sigma > 1$ and $\eta > \alpha$, fixed-duration expansions of research compute make the truncated value unbounded above. At $\eta = \alpha$, the leading benefit and cost terms have the same order, so their coefficients determine the result. For a fixed finite frontier, the truncated problem instead has a finite value and attains its maximum for every $0 < \eta < 1$.*

The uncapped result compares the cost of research with its effect on future operating profit. Expenditure first raises AI efficiency; higher AI efficiency then raises the profit from supplying effective AI production services. The proof bounds these two effects in turn and compares their combined growth with the linear cost of research.

Proof of Proposition 5. I first consider the uncapped economy. Define

$$\theta = \frac{1-\alpha}{\alpha}, \qquad S = \int_0^T M_t \, \mathrm{d}t.$$

Here S is total research expenditure over the fixed horizon $[0, T]$. For an admissible research policy, let its discounted net payoff be

$$J_T(M) = \int_0^T \exp\left\{-\int_0^t r_s \, \mathrm{d}s\right\} [\pi(B_t) - M_t] \, \mathrm{d}t.$$

For every admissible policy, B is nondecreasing. The AI-efficiency law implies

$$\frac{\mathrm{d}}{\mathrm{d}t} B_t^{1-\eta} = (1-\eta)\chi M_t^\eta.$$

Integration and Hölder's inequality give

$$B_t^{1-\eta} = B_0^{1-\eta} + (1-\eta)\chi \int_0^t M_s^\eta \, \mathrm{d}s,$$

$$\int_0^T M_t^\eta \, \mathrm{d}t \le T^{1-\eta} S^\eta.$$

Because $0 < \eta < 1$, spreading a given expenditure evenly over $[0, T]$ maximizes the integral of M^η. Hence there is a constant $C_B > 0$, independent of the research policy, such that

$$\max_{0 \le t \le T} B_t \le C_B \left(1 + S^{\eta/(1-\eta)}\right), \tag{64}$$

The same bound covers arbitrarily concentrated controls because, on a support of length $d \le T$, $\int M_t^\eta \, \mathrm{d}t \le d^{1-\eta} S^\eta$.

Let $\overline{K} = \max_{t \le T} K_t$ and $\overline{\mathcal{L}} = \max_{t \le T} A_t L_t$. The weighted power-mean inequality gives $Z \le AL + X$. The final-good firm's demand condition therefore implies

$$p_X X \le (1-\alpha)\overline{K}^\alpha \left(\overline{\mathcal{L}}^{1-\alpha} + X^{1-\alpha}\right).$$

Write the right-hand side as $a + bX^{1-\alpha}$, with $a, b > 0$ determined by the given paths and production parameters. The first-order condition for maximizing this revenue bound net of X/B is $b(1-\alpha)X^{-\alpha} = 1/B$, and its maximized value is

$$\max_{X \ge 0} \left\{ a + bX^{1-\alpha} - \frac{X}{B} \right\} = a + \alpha b [b(1-\alpha)B]^\theta.$$

There is therefore a constant $C_\pi > 0$, independent of the research policy, such that

$$\pi(B) \le C_\pi (1 + B^\theta). \tag{65}$$

Combining the AI-efficiency and profit bounds shows that operating profit grows no faster than a constant plus a term proportional to

$$S^{\eta(1-\alpha)/[\alpha(1-\eta)]}, \tag{66}$$

where the exponent is the product of $\eta/(1-\eta)$, from expenditure to AI efficiency, and $(1-\alpha)/\alpha$, from AI efficiency to profit. This is an upper bound on the order of growth, not an exact profit function.

On a finite horizon, integrability of r bounds the discount factor above and away from zero. Let $\underline{D} > 0$ be its minimum on $[0, T]$. Combining (64) and (65) yields, for constants $C_0, C_1 > 0$ independent of the research policy,

$$J_T(M) \le C_0 + C_1 S^p - \underline{D} S, \qquad p = \frac{\eta(1-\alpha)}{\alpha(1-\eta)}. \tag{67}$$

The exponent satisfies $p < 1$ exactly when $\eta < \alpha$. In this case, the benefit grows at most sublinearly in expenditure while its cost grows linearly. The bound tends to minus infinity as $S \to \infty$, so the value is finite and the objective is coercive.

An upper bound with $p > 1$ does not by itself prove unboundedness. For the converse, I construct research policies with unbounded payoffs. Suppose $\sigma > 1$. As $B \to \infty$, optimized operating profit satisfies

$$\pi(B) \sim \kappa_X K B^{\theta}$$

for a constant $\kappa_X > 0$, uniformly when K, A, and L remain in compact subsets of the positive orthant. Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. Then

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1-\eta)\chi \mathcal{M}^{\eta} \min\{t, \Delta\}.$$

Choosing X optimally at every date gives

$$J_T(\mathcal{M}) = c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M} + o\Big(\mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]}\Big), \tag{68}$$

with $c_\pi, c_M > 0$. The exponent exceeds one when $\eta > \alpha$, so this family makes the truncated value unbounded above. At equality, the exponent is one and the sign of the leading coefficient determines this family's result.

For a finite frontier, the invariance result above confines B to the compact interval $[B_0, \overline{B}]$. Optimized operating profit is continuous in B and uniformly bounded on the fixed horizon. Hence $J_T(M) \leq C_0 - \underline{D}S$, so every maximizing sequence has bounded total expenditure. Set $v = M^{\eta}$ and $p = 1/\eta > 1$. The control cost is v^p, while the state equation is affine in v:

$$\dot{B} = \chi B^{\eta} \left(1 - \frac{B}{\overline{B}}\right) v.$$

A maximizing sequence is bounded in the reflexive space $L^p[0, T]$. A weakly convergent subsequence and the associated uniformly convergent state paths preserve feasibility. The operating-profit integral is continuous under uniform state convergence, while $\int v^p$ is weakly lower semicontinuous. The objective is therefore weakly upper semicontinuous and attains its maximum. $\square$

The condition $\eta < \alpha$ prevents arbitrarily large research expenditure from being optimal on a fixed horizon in the uncapped economy. It is a sufficient restriction, not a necessary one for every value of σ, and finite-horizon existence with a finite frontier does not require $\eta < \alpha$. Neither the bound nor finite-horizon developer existence establishes an equilibrium: the paths of K, A, L, and r were taken as given throughout this argument. The quantitative analysis uses $\eta = 0.20 < \alpha = 0.33$, for which the expenditure exponent is approximately 0.508; the value of η is illustrative rather than estimated.

Global developer verification. For a finite frontier, define research depth as the monotone transformation

$$R_B(B) = -\overline{B}\log\left(1-\frac{B}{\overline{B}}\right), \qquad B(R_B) = \overline{B}\left(1-e^{-R_B/\overline{B}}\right). \tag{69}$$

It maps $B \in [B_0, \overline{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$, and rewrites the AI-efficiency law as

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \tag{70}$$

Write $\pi_t(B)$ for optimized operating profit evaluated at the date-t values of K, A, and L, and define the shadow value of research depth as $q_{R,t} = q_t\psi(B_t)$. Holding this shadow value fixed, let

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) \equiv \max_{\widetilde{M}\geq 0}\left\{\pi_t(B(\widetilde{R}_B)) - \widetilde{M} + q_{R,t}\chi[B(\widetilde{R}_B)\widetilde{M}]^\eta\right\}.$$

Lemma 2 (Global verification of the developer's research path)**.** *Let $0 < \eta < 1$ and $\chi > 0$. Fix $0 < B_0 < \overline{B} < \infty$ and positive continuous paths for K, A, and L, and take the locally integrable interest-rate path r as given. Consider a feasible candidate $\{B_t, M_t\}_{t\geq 0}$ for problem (21), with continuously differentiable B_t, interior research $M_t > 0$, and a finite objective. Suppose there is a continuously differentiable costate q_t satisfying the research condition (22), the costate equation (23), and the transversality condition (25). If, for every $t \geq 0$ and every $\widetilde{R}_B \geq R_{B,0}$,*

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) \leq \widehat{\mathcal{H}}_t(R_{B,t}) + \widehat{\mathcal{H}}_t'(R_{B,t})(\widetilde{R}_B - R_{B,t}), \tag{71}$$

then the candidate is a global maximizer of the reduced developer problem. Admissible alternatives have $M_t \geq 0$, locally finite research expenditure, and the same initial state; the comparison also bounds the upper limit of their truncated discounted payoffs. Together with the unique static service choice characterized in Lemma 1, it is globally optimal for the developer.

Proof of Lemma 2. Fix a date and the candidate shadow price $q_{R,t} = q_t\psi(B_t) > 0$. Maximizing the research-depth Hamiltonian over research expenditure gives

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) = \pi_t(B(\widetilde{R}_B)) + \frac{1-\eta}{\eta}M_t\left[\frac{B(\widetilde{R}_B)}{B_t}\right]^{\eta/(1-\eta)}. \tag{72}$$

Here M_t is the candidate's research expenditure. The formula follows by maximizing $-M + q_{R,t}\chi[B(\widetilde{R}_B)M]^\eta$ over $M \geq 0$ and using the candidate's research first-order condition. Its derivative at the candidate state is

$$\widehat{\mathcal{H}}_t'(R_{B,t}) = \frac{\psi(B_t)}{B_t}(U_t + M_t). \tag{73}$$

Write $D_t = \exp(-\int_0^t r_s\,\mathrm{d}s)$ and $\mu_t = D_t q_{R,t}$. The transformed costate equation is $\dot{\mu}_t = -D_t\widehat{\mathcal{H}}_t'(R_{B,t})$. For any admissible alternative research policy and its state $\widetilde{R}_{B,t}$, maximizing over its control first bounds its Hamiltonian by (72). Apply (71), subtract the candidate payoff, and integrate by parts. Equal initial states imply

$$\widetilde{J}_T - J_T \leq -\mu_T(\widetilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T}.$$

Because

$$\psi(B_t)R_{B,t} = -(\overline{B} - B_t)\log\left(1 - \frac{B_t}{\overline{B}}\right) \leq B_t,$$

the last expression converges to zero by the developer's transversality condition. Taking the upper limit establishes global optimality. The argument requires a global upper tangent at the candidate state, not concavity at every counterfactual state. □

A concavity test for Lemma 2. Fix a date t and an alternative AI-efficiency level $b \in [B_0, \overline{B})$. Let $X_t^*(b)$ be the unique maximizer of problem (18), evaluated at K_t, A_t, and L_t, and let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding values. Implicitly differentiating the monopoly condition (19), using Equations (10) and (15), gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \tag{74}$$

This endogenous elasticity is not an additional model parameter. In what follows, suppress the date and counterfactual-state arguments and write

$$X^*(B) \equiv X_t^*(b), \qquad \varepsilon_B \equiv \varepsilon_{B,t}(b).$$

Because $\pi_B = X^*(B)/B^2$,

$$\pi_{BB} = \frac{X^*}{B^3}(\varepsilon_B - 2). \tag{75}$$

A direct differentiation gives

$$\frac{\partial^2 \pi(B(R_B))}{\partial R_B^2} = \frac{X^*(B)\psi(B)}{B^3}\left[\psi(B)(\varepsilon_B - 2) - \frac{B}{\overline{B}}\right]. \tag{76}$$

The transformation (69) satisfies

$$B'(R_B) = \psi(B) > 0, \qquad B''(R_B) = -\frac{\psi(B)}{\overline{B}} < 0.$$

Using $\pi_B = X^*(B)/B^2$ and (75) gives (76). Hence the algebraic condition

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\overline{B}} \tag{77}$$

for every $t \geq 0$ and $b \in [B_0, \overline{B})$ makes optimized operating profit concave in research depth throughout the reachable state domain.

The research term in (72) is a positive multiple of $B(R_B)^\mu$, where $\mu = \eta/(1-\eta) > 0$. Its curvature is

$$\frac{\mathrm{d}^2 B(R_B)^\mu}{\mathrm{d}R_B^2} = \mu B^{\mu-2}\psi(B)\left[(\mu-1)\psi(B) - \frac{B}{\overline{B}}\right]. \tag{78}$$

Thus (77), together with

$$\frac{B_0}{\overline{B}} \geq \max\left\{0, \frac{2\eta - 1}{\eta}\right\}, \tag{79}$$

makes the maximized Hamiltonian concave throughout the reachable state domain and proves (71). If $\eta \leq 1/2$, condition (79) holds automatically. But for every fixed $0 < \eta < 1$ its right-hand side is strictly below one, so it also holds without this restriction when initial efficiency is sufficiently close to the frontier. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state.

At unit elasticity, the concavity test has a simple primitive form. Define $\beta = (1-\alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$. Because $B(R_B)$ is increasing and concave, $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. In the benchmark calibration in Table 2, $\beta = 0.134$, so this sufficient condition holds with substantial slack. It is not imposed as an additional maintained assumption; outside this range, the support condition can instead be checked directly.

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity defined in Equation (74) satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b. Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X)\left[(1-\sigma^{-1})(1-s_X) + (\sigma^{-1} - \alpha)s_X\right] \geq 0.$$

Choose $B_* \in (B_0, \overline{B})$ with $B_*/\overline{B} > \max\{0, (1-2\alpha)/(1-\alpha)\}$. Operating profit is then concave in research depth for $B \geq B_*$. If $\eta \leq 1/2$, the research term in (72) is concave there as well. For a candidate with $B_t \geq B_*$, it consequently suffices to check

$$\widehat{\mathcal{H}}_t(R_{B,t}) - \widehat{\mathcal{H}}_t'(R_{B,t})(R_{B,t} - R_{B,0}) \geq \widehat{\mathcal{H}}_t(R_B(B_*)). \tag{80}$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B, so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (80), divided by Y_t, converges to

$$\alpha(1-\alpha)\left[1 - \left(\frac{B_*}{\overline{B}}\right)^{(1-\alpha)/\alpha}\right] > 0.$$

To obtain this limit, use $M/Y \to 0$, $\psi(B)R_B \to 0$, and $\pi(B_*)/Y \to \alpha(1-\alpha)(B_*/\overline{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

Preliminary results for Propositions 1–3. I first construct the terminal values, then a family of convergent solutions of the exact equilibrium equations, and finally verify optimality and transversality. No equilibrium trajectory or convergence assumption is used as a premise.

Write $\theta = (1-\alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma-1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \qquad \mathcal{H}_\sigma(s) = \left[(1-\alpha) s\, m(s) \left(\frac{\omega_X}{s}\right)^{1/\varphi}\right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X(1-s_X)}\right]^{1/\varphi}, \qquad \frac{U}{Y} = (1-\alpha) s_X m(s_X), \tag{81}$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \tag{82}$$

For example, $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1-\alpha)s_X m(s_X) Y$ give the last identity directly.

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1-\sigma}{1-\alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1-\alpha)\omega_X^{1/\varphi} m(s) s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1-\alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho+\gamma+\delta}{\alpha} = \overline{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \tag{83}$$

has exactly one interior solution if and only if $\overline{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0,1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma-2)s - \varphi$. If $a(\sigma-2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma-2) < (\sigma-2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma-2) < (1-\sigma^{-1})(2-\sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (83) has exactly one interior solution if and only if $\overline{B} < \mathcal{B}(\sigma)$. These calculations also derive (32)–(33).

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN, respectively.

Construct the normalized terminal values sequentially:

$$\begin{aligned}
\overline{x} &= \left[\frac{\omega_L \overline{s}_X}{\omega_X(1-\overline{s}_X)}\right]^{1/\varphi}, & \overline{u} &= \frac{\overline{x}}{\overline{B}}, \\
\overline{y} &= \frac{\overline{u}}{(1-\alpha)\overline{s}_X m(\overline{s}_X)}, & \overline{k} &= \frac{\alpha\overline{y}}{\rho+\gamma+\delta}, \\
\overline{c} &= \overline{y}-\overline{u}-(\delta+n+\gamma)\overline{k}, & \overline{r} &= \rho+\gamma.
\end{aligned} \tag{84}$$

At $\sigma = 1$, define $\beta = (1-\alpha)\omega_X$ as in the main text. The exact static solution is

$$y = \left[(\beta^2 B)^{\beta} k^{\alpha}\right]^{1/(1-\beta)}, \qquad u = \beta^2 y, \qquad x = Bu.$$

The equation $y/k = (\rho+\gamma+\delta)/\alpha$ gives (36). It has exactly one positive solution for every finite $\overline{B} > 0$; set $\overline{y} = [(\rho+\gamma+\delta)/\alpha]\overline{k}$, $\overline{u} = \beta^2\overline{y}$, and $\overline{x} = \overline{B}\,\overline{u}$, with $\overline{c}, \overline{r}$ as in (84).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\overline{s}_X) < 1$. Since $0 < \overline{s}_X < 1$, in all three cases

$$\frac{\overline{u}}{\overline{y}} = (1-\alpha)\overline{s}_X m(\overline{s}_X) < 1-\alpha, \qquad \frac{(\delta+n+\gamma)\overline{k}}{\overline{y}} = \frac{\alpha(\delta+n+\gamma)}{\rho+\gamma+\delta} < \alpha.$$

It follows that $\overline{c}/\overline{y} > 0$.

Common positive-labor-share construction. Use the state coordinates k, d, τ introduced before Proposition 1, together with consumption and the AI-efficiency shadow price divided by AN:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\overline{B}-B)AN, \quad \widetilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \tag{85}$$

They leave the model unchanged. Given k, B, solve the unique static monopoly problem for y, x, u, and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\overline{B}-B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \overline{B} - d\tau, \qquad M = \left[\widetilde{q} d\,\chi\eta\frac{B^{\eta}}{\overline{B}}\right]^{1/(1-\eta)}, \qquad a = \chi\frac{B^{\eta}M^{\eta}}{\overline{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \tag{86a}$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \tag{86b}$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \tag{86c}$$

$$\frac{\dot{\widetilde{q}}}{\widetilde{q}} = r - \frac{x}{\widetilde{q}B^2} - \eta\frac{ad\tau}{B} + a - n - \gamma, \tag{86d}$$

$$\dot{\tau} = -(n+\gamma)\tau. \tag{86e}$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the terminal construction by setting

$$\overline{M} = \left[\frac{(n+\gamma)\overline{B}^{1-\eta}}{\chi}\right]^{1/\eta}, \qquad \overline{\widetilde{q}} = \frac{\overline{x}}{\overline{r}\,\overline{B}^2}, \qquad \overline{d} = \frac{\overline{M}}{\eta(n+\gamma)\overline{\widetilde{q}}}. \tag{87}$$

All are positive. Direct substitution shows that $(\overline{k}, \overline{c}, \overline{d}, \overline{\widetilde{q}}, 0)$ is a fixed point of (86). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \overline{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1-\eta)$, $r_k = \partial r/\partial \log k$, and $a_k = \partial[(y-u)/k]/\partial \log k + \overline{c}/\overline{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \widetilde{q}, \tau)$ coordinates the Jacobian has the two diagonal blocks

$$J_R = \begin{pmatrix} a_k & -\overline{c}/\overline{k} \\ r_k & 0 \end{pmatrix}, \qquad J_B = \begin{pmatrix} -(n+\gamma)\mu & -(n+\gamma)\mu \\ (n+\gamma)\mu & \overline{r} + (n+\gamma)\mu \end{pmatrix}, \tag{88}$$

and the additional eigenvalue $-(n+\gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \widetilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression in (62). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1-e)}{D(s)}, \qquad r_k = (r+\delta)(1-\alpha)\left[\frac{\alpha s(1-e)}{D(s)} - 1\right].$$

Moreover,

$$D(s) - \alpha s(1-e) = (1-s)\{u_\sigma(1-e) + (\alpha - u_\sigma)(1-u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s, with positive endpoint values $u_\sigma(1-u_\sigma)$ and $u_\sigma^2 + \alpha(1-2u_\sigma)$. The latter is positive if $u_\sigma \leq 1/2$; otherwise $\alpha < u_\sigma$ implies it exceeds $u_\sigma(1-u_\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \qquad \det J_B = -(n+\gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. To verify that their number suffices to select the jump variables, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The homogeneous Ramsey block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\widetilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \overline{B} - d\tau$ remains positive nearby. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \widetilde{q})$ over (k, d, τ); see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$. Every physical initial state satisfying (34) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \widetilde{q}, k$ and the original AI-efficiency law ensure $0 < B_t < \overline{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (35) and (37). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of its smooth static ratios tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

AI-dominated construction. Let $\sigma > 1$ and $\overline{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. Use the alternative coordinates

$$h = (AN/K)^{\varphi}, \quad d = (\overline{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. Then

$$B = \overline{B} - d\tau h^{1/\varphi}, \quad M = \left[vd\,\chi\eta B^{\eta}/\overline{B}\right]^{1/(1-\eta)}, \quad a = \chi B^{\eta} M^{\eta}/\overline{B}, \quad r = \alpha y - \delta.$$

Define $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{89a}$$

$$\dot{d}/d = g_K - a, \tag{89b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{89c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta a d\tau h^{1/\varphi}/B + a - g_K, \tag{89d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{89e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^{\varphi}]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1-\alpha)$. Its implicit static solution is therefore smooth in (h, B) near $(0, \overline{B})$. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. The C^1 stable-manifold theorem applies; see Feldman (2021, p. 7). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned}
&\overline{y} = \overline{\zeta}, && \overline{u} = (1-\alpha)^2\overline{y}, && \overline{x} = \overline{B}\,\overline{u}, \\
&\overline{r} = \alpha\overline{y} - \delta, && \overline{g} = n + \overline{r} - \rho, && \overline{c} = \alpha(1-\alpha)\overline{y} + \rho - n > 0, \\
&\overline{v} = \overline{x}/(\overline{r}\,\overline{B}^2), && \overline{M} = \left[\overline{g}\,\overline{B}^{1-\eta}/\chi\right]^{1/\eta}, && \overline{d} = \overline{M}/(\eta\overline{g}\,\overline{v}).
\end{aligned} \tag{90}$$

The threshold implies $\overline{g} > n + \gamma > 0$ and $\overline{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \overline{d}, \overline{c}, \overline{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \overline{g}) < 0$ and $-(n + \gamma) < 0$. The consumption-ratio eigenvalue is $\overline{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\overline{g}\mu & -\overline{g}\mu \\ \overline{g}\mu & \overline{r} + \overline{g}\mu \end{pmatrix}, \qquad \det = -\overline{g}\mu\overline{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects invertibly onto (h, d, τ). The same graph argument constructs the family specified by (39), with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios Y/K, X/K, and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \to \overline{g}$. Also $s_X \to 1$ and $1 - s_X$ is asymptotically proportional to $(AN/X)^{\varphi}$. The wage equation gives

$$g_w \to \overline{g} - n + \varphi(n + \gamma - \overline{g}) = \gamma + \frac{\overline{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 3.

Optimality and transversality. In either construction let $\overline{g}$ denote the terminal aggregate growth rate: $n+\gamma$ in the positive labor-share case and $n+\overline{r}-\rho$ in the AI-dominated case. The constructions imply

$$a \to \overline{g}, \qquad M \to \left[\frac{\overline{g}\,\overline{B}^{1-\eta}}{\chi}\right]^{1/\eta} > 0, \qquad g_q \to \overline{g}, \qquad g_B \to 0, \qquad M/Y \to 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \to n - \rho < 0, \qquad -r + g_q + g_B \to \overline{g} - \overline{r} = n - \rho < 0. \tag{91}$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, the household budget is linear at given prices and distributions. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log utility and equal initial capital give, for any admissible alternative $(\widetilde{C}, \widetilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\widetilde{C}_t}{C_t}\,\mathrm{d}t \le \int_0^T \lambda_t(\widetilde{C}_t - C_t)\,\mathrm{d}t = -\lambda_T(\widetilde{K}_T - K_T) \le \lambda_T K_T \to 0.$$

Here $\widetilde{K}_T \ge 0$ and the last limit is the household TVC. Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency argument. Let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s\,\mathrm{d}s)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \le \pi_t(b) \le C_\pi S_t$ for every date and every $b \in [B_0, \overline{B}]$. Since $\mathrm{d}\log(D_t S_t)/\mathrm{d}t \to \overline{g} - \overline{r} = n - \rho < 0$, $\int_0^\infty D_t S_t\,\mathrm{d}t < \infty$. Thus discounted operating profit is uniformly bounded above across all alternative research policies. The candidate's objective is finite, because its research compute is bounded and $r_t \to \overline{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions (77) and (79) for every date and every alternative AI-efficiency level in $[B_0, \overline{B})$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of counterfactual efficiencies $[b_*, \overline{B}]$, with $0 < b_* < \overline{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\overline{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\overline{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \to 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \le E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1-\eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \overline{B})$

sufficiently close to $\overline{B}$ that

$$(1 - b_{**}/\overline{B}) \max\{E, \mu - 1, 0\} < b_{**}/\overline{B}.$$

Shrink $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 \geq b_{**}$. This is possible because $B_0 = \overline{B} - d_0\tau_0$ in the first construction and $B_0 = \overline{B} - d_0\tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 2 and the verified TVC therefore establish global developer optimality without imposing $\eta \leq 1/2$. No curvature property of the candidate is being assumed: it follows from the primitives and the chosen initial-state neighborhood.

All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore equilibria, not merely solutions of the canonical equations. The result is local only in its set of initial stocks. For $\sigma > 1$, at $\overline{B} = \mathcal{B}(\sigma)$ the interior-share terminal point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\overline{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit.

Proof of Proposition 1. For $0 < \sigma < 1$ and $\overline{B} > \mathcal{B}(\sigma)$, the share map above has a unique interior root. Equations (84) and (87) construct the positive terminal values. The common positive-labor-share construction gives the local stable manifold and the unique nearby choices of C_0 and q_0. The optimality and transversality argument then verifies that these paths satisfy Definition 1, with the limits stated in (35). □

Proof of Proposition 2. At $\sigma = 1$, the exact static solution above determines positive terminal values for every finite $\overline{B} > 0$. The common positive-labor-share construction again gives the local stable manifold and the unique nearby choices of C_0 and q_0. The optimality and transversality argument verifies Definition 1 and yields (37). □

Proof of Proposition 3. For $\sigma > 1$, the share map above is strictly decreasing. If $\overline{B} < \mathcal{B}(\sigma)$, it has a unique interior root and the common positive-labor-share construction applies. If $\overline{B} > \mathcal{B}(\sigma)$, the AI-dominated construction gives the local stable manifold, the unique nearby choices of C_0 and q_0, and the limits in (40) and (41). The common optimality and transversality argument verifies Definition 1 in both cases. At $\overline{B} = \mathcal{B}(\sigma)$, the boundary fixed point is not hyperbolic, so the argument does not apply. □

Proof of Proposition 4. Propositions 1 and 2, together with the below-threshold case of Proposition 3, imply $g_w \to \gamma$ whenever the limiting labor share is positive. The above-threshold case of Proposition 3 implies both $g_w \to \gamma + (\overline{r} - \rho - \gamma)/\sigma > \gamma$ and $wL/Y \to 0$. This proves the equivalence. Subtract the wage-growth limit in (41) from the per-person output-growth limit in (40). Moreover, $\mathrm{d}\log(wL/Y)/\mathrm{d}t = g_w + n - g_Y$ because $L = N$. Finally, combine (43) with (41). □

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and algorithm in words

The numerical problem is to select an equilibrium path, not merely to integrate the laws of motion from four arbitrary initial values. Capital and AI efficiency are predetermined at date zero, whereas consumption and the developer's shadow value can jump. For given K_0 and B_0, almost every choice of C_0 and q_0 sends the economy away from the terminal equilibrium regime or violates feasibility. The algorithm therefore chooses C_0 and q_0 jointly with the entire path so that the economy starts from the prescribed stocks and approaches the appropriate stable continuation.

The calculation proceeds in six steps:

1. For each value of σ, I use the analytical results in Section 4 to identify the relevant terminal regime and its normalized limiting point.

2. Near that point, I construct a local boundary-value solution. The terminal restrictions eliminate the two unstable directions rather than fixing arbitrary terminal levels for the jump variables.

3. Conditional on every trial value of the dynamic variables, I solve the static monopoly condition for AI production services and recover research expenditure from the developer's first-order condition. Thus the numerical system uses the static choices already derived in Section 3; it does not add behavioral equations.

4. I use continuation to move the local solution gradually to the common prescribed values of K_0, B_0, and A_0N_0. Each successful solution is the initial guess for the next continuation stage. This step selects the two date-zero jump variables without treating the no-AI economy as a mechanical starting point.

5. I lengthen the terminal horizon and tighten the solver tolerances. A valid approximation should leave the date-zero jumps and the entire displayed path essentially unchanged.

6. I admit the path only after separate checks of feasibility, the original equilibrium equations, static optimality, global developer optimality, both transversality conditions, and consistency with the analytical terminal continuation. A converged boundary-value calculation that fails any of these checks is not reported as an equilibrium.

This procedure can be viewed as finding the stable saddle path from both ends. The initial boundary supplies the predetermined stocks, while the terminal boundary rules out explosive components. Collocation chooses the whole path at once and, with it, the initial consumption and shadow value that reconcile the two boundaries. The terminal date is a computational truncation; the infinite-horizon interpretation comes from the analytical continuation and the transversality checks described below.

B.2 Notation in the paper and code

The numerical code retains its existing variable names. To make the implementation auditable, the notation used in the paper maps into the code as follows:

$$\gamma = \texttt{labor\_productivity\_growth}, \qquad \sigma = \texttt{sigma\_xl}, \qquad \psi(B) = \exp\bigl(\texttt{log\_psi}\bigr). \tag{92}$$

This map changes only the notation in the exposition. It does not redefine a variable or parameter, alter its timing, or modify any equation used by the simulations.

B.3 Boundary-value formulation

A boundary-value problem (BVP) selects the initial consumption and shadow price jointly with the subsequent path. I use the existing collocation solver, `scipy.integrate.solve_bvp`, for the four dynamic equations in the equilibrium definition. At every evaluation, I solve the static monopoly condition and recover research expenditure from its first-order condition. Effective labor is exogenous, $AL = A_0 N_0 e^{(n+\gamma)t}$, and therefore is not an additional unknown path in this dated BVP.

The computational coordinates are

$$\left(\log K,\ \log \frac{B}{\overline{B} - B},\ \log C,\ \log q\right).$$

They enforce positivity and $0 < B < \overline{B}$ without clipping a computed equilibrium. I subtract the analytical terminal aggregate growth rate times t from each coordinate to remove its linear trend. This is a numerical transformation, not a change in the model's variables. In particular, the code stores $\log \psi(B)$ separately: a displayed ratio $B/\overline{B}$ can round to one even though the remaining frontier gap is strictly positive.

The left boundary fixes only K_0 and B_0: from (52) in the main comparison and from (57) in the Ramsey-start experiment, (58)– (59) in the near-terminal diagnostic, and (60) in the slow-transition exercise. At the right boundary, the solution is mapped into the regime-specific normalized coordinates of Appendix A. Two projection conditions remove the unstable components of the linearized terminal system. Its third stable direction is associated with the exogenous scale. The two stock conditions and two terminal projections thus close a four-dimensional problem; neither C_0 nor q_0 is predetermined. A finite terminal projection approximates the stable continuation, which is why increasing the horizon is a necessary numerical check.

For $\sigma \leq 1$, I use the positive-labor-share terminal formulas. For $\sigma > 1$, the frontier's position relative to $\mathcal{B}(\sigma)$ selects the terminal regime. At $\sigma = 1$, the CES block uses its exact Cobb–Douglas limit. Near one, `log1p` and `expm1` evaluate the CES expression without cancellation, and the frontier comparison is performed in logarithms. I solve the labor-supported terminal static condition directly for $\log[X/(AL)]$, rather than invert a share formula by dividing by $\sigma - 1$. A trial quantity with nonpositive marginal revenue can bracket the monopoly optimum, but is never accepted as that optimum.

B.4 Continuation and numerical accuracy

For each elasticity, I first solve a local BVP near its analytical terminal point. I then move the initial stocks and effective-labor scale continuously to their prescribed values, retaining the preceding solution as the next guess. Each design uses 32 continuation stages. There is no continuation through the degenerate no-AI economy. The common main stocks are those in (52); the uncapped unit-elastic reference supplies stocks for the slow-transition and financing sensitivities, but never supplies their jump variables.

The initial horizon combines the time needed to reach the terminal scale with eleven decay times of the slowest stable eigenvalue. This gives an initial truncation criterion, not an equilibrium assumption. I extend that horizon twice by 500 years, tightening the collocation tolerance from 2×10^{-6} to 10^{-8} and then 10^{-9}. The final boundary tolerance is 10^{-11}. Initial meshes of 221, 401, and 601 points are refined adaptively by the solver. The final horizons exceed the 500-year display windows and the slow exercise's 4,000-year window, as applicable; the figures contain no extrapolated spline values.

Table 4: Numerical accuracy of the four displayed equilibria

σ	Final horizon	Mesh points	Dynamic residual	Horizon change
0.90	4,946.2	874	9.17×10^{-10}	3.26×10^{-8}
1.00	4,821.1	872	1.05×10^{-9}	3.73×10^{-8}
1.10	4,737.0	909	9.12×10^{-10}	3.37×10^{-8}
1.50	3,008.5	1,092	8.19×10^{-10}	1.56×10^{-8}

Dynamic residuals reconstruct the four original equations using five-point differences of spline values at 1,001 dates, with steps of 0.003 and 0.001 years; the table reports the larger maximum. Horizon change is the largest change in the detrended logarithmic coordinates over years 0–500 between the last two solves. Values are rounded upward.

The slower-transition exercise repeats this entire calculation rather than rescaling or extending the main trajectories. Its final horizons are 6,857.1, 6,731.9, 6,647.9, and 4,919.4 years for $\sigma = 0.90, 1, 1.10, 1.50$, respectively. The corresponding independently reconstructed dynamic residuals are below 1.1×10^{-9}, and extending the horizon changes the detrended coordinates by at most 2.8×10^{-8} over the full 0–4,000 display window.

The Ramsey-start experiment also repeats the complete calculation for four separate BVPs. Its final horizons are 4,946.2, 4,821.1, 4,737.0, and 3,008.5 years for $\sigma = 0.90, 1, 1.10, 1.50$. The independently reconstructed dynamic residuals are below 1.1×10^{-9}, and the final horizon extension changes the detrended coordinates by at most 3.8×10^{-8} over years 0–500.

The maximum research and static monopoly first-order residuals are below 1.8×10^{-13}. Initial jump choices change by less than 1.8×10^{-10} in logs in the final horizon comparison. The normalized terminal coordinates differ from their analytical limits by less than 2.3×10^{-6}. The acceptance thresholds are deliberately looser than these observed errors: 10^{-6} for independently reconstructed dynamic residuals, 10^{-9} for static first-order conditions, 2×10^{-5} for horizon sensitivity, and 10^{-4} for terminal-coordinate gaps. Tightening the solver and extending its horizon provide the evidence of stability; the thresholds do not replace any economic condition.

B.5 Optimality and the infinite-horizon continuation

The equation audit also checks feasibility, the monopoly second-order condition, and that numerical safeguards used for off-path Newton trials do not bind on the solution. Both transversality expressions converge to zero on each selected continuation: their asymptotic log growth rate is $n - \rho = -0.037$. This analytical rate, rather than a small terminal value alone, establishes the transversality check.

For $\sigma = 0.90, 1, 1.10$, the developer's operating profit is concave in research depth over the counterfactual AI-efficiency domain checked at each date. The final audit uses 161 dates and 161 AI-efficiency points per date, with scalar minimization between grid points. For $\sigma = 1.50$, this stronger curvature test fails at low counterfactual AI-efficiency levels; checking concavity only along the realized path would miss that failure.

I instead apply the support condition in Lemma 2. At each date, I minimize its tangent gap, divided by the candidate's Y, over counterfactual AI-efficiency levels $B \geq B_0$, refining every local minimum found on the grid. An analytical upper bound on the Hamiltonian's derivative covers the unbounded research-depth domain beyond the numerical grid. The check is repeated with 81 dates and 101 AI-efficiency points, then with 321 dates and 241 points. In the main, Ramsey-start, and slow-transition comparisons, the smallest normalized gaps are of order -10^{-14}, consistent with floating-point error at the tangent point. The numerical gap tolerance is 10^{-10}; the economic support inequality remains exact.

For the infinite-horizon continuation, I use (80) with $B_* = 0.75\overline{B}$. Its limiting margin divided by output is approximately $0.09781 > 0$; the computed terminal margin differs from that limit by less than 6×10^{-12}. Thus the analytical continuation satisfies the sufficient support condition eventually, while the dated audit checks the computed transition. These are numerically verified equilibrium approximations supported by analytical terminal and optimality results, not interval-arithmetic existence certificates for the common initial stocks. Nor do they establish existence for every initial condition or every parameter vector.

The financing sensitivity reported in Section 5.7 is solved separately rather than spliced onto a main trajectory. It uses $K_0 = 2.027733654$, $B_0 = 0.443670932$, and $\chi = 1.4223$, preserving the 50-year transition midpoint while changing the predetermined stocks. The calculation repeats both horizon extensions, the independent equation and TVC checks, and the two Hamiltonian-support grids. It passes the same admission thresholds as the main $\sigma = 1.50$ path.

The slow-transition comparison in Section 5.6 likewise uses separate BVPs, checkpoints, audit files, and manifests. It sets $\chi = 0.01$ and uses the earlier stocks in (60); all four paths pass the equation, horizon, TVC, and developer-optimality gates before export.

The Ramsey-start comparison in Section 5.4 begins directly on the positive-AI branch with the predetermined stocks in (57). The numerical continuation changes those initial stocks, not ω_X: it never passes through the degenerate no-AI problem. All four paths pass the same equation, horizon, TVC, and developer-optimality gates before export.

B.6 Reproduction and export safeguards

The implementation preserves the earlier solver. The entry point is `scripts/simulate_rewrite_finite_frontier.py`; the final audit is `scripts/audit_rewrite_equilibria.py`. The reproducible sequence is: solve each of the four elasticities with `--sigma`; run `--verify-long-horizon`; run `audit_rewrite_hamiltonian_support.py` for $\sigma = 1.50$ at both grid resolutions; run the final audit; run `audit_initial_financing_sensitivity.py`; export; and render with `plot_rewrite_equilibria.py`. The public reproduction command performs that sequence for the main, Ramsey-start, slow, and near-terminal designs, using display horizons of 500 years except for the 4,000-year slow exercise. All code, audit files, plot-ready data, and figures used in these simulations are available in the paper's public GitHub repository. The repository's replication guide provides a one-command workflow, installation instructions, an output map, and the criteria used to interpret the diagnostics. The execution record `audit/rewrite_equilibrium_simulations.md` reports the commands and test results for the version used in the paper.

The exported data and figures require all four scenarios to pass admission. Checkpoint hashes link each audit to its solved spline; a second manifest links the plotted CSV to those checkpoints. Failed or stale candidates cannot silently enter a partial comparison. The data retain the original model variables and use only the normalizations stated in the figure captions.

References

Acemoglu, D. and Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30.

Aghion, P. and Howitt, P. (1992). A model of growth through creative destruction. *Econometrica*, 60(2):323–351.

Aghion, P., Jones, B. F., and Jones, C. I. (2019). Artificial intelligence and economic growth. In Agrawal, A., Gans, J., and Goldfarb, A., editors, *The Economics of Artificial Intelligence: An Agenda*, pages 237–282. University of Chicago Press.

Athey, S. and Scott Morton, F. (2025). Artificial intelligence, competition, and welfare. NBER Working Paper 34444, National Bureau of Economic Research.

Autor, D. H. (2013). The "task approach" to labor markets: An overview. *Journal for Labour Market Research*, 46(3):185–199.

Autor, D. H. and Kausik, B. N. (2026). Resolving the automation paradox: Falling labor share, rising wages. arXiv:2601.06343.

Bloom, N., Jones, C. I., Van Reenen, J., and Webb, M. (2020). Are ideas getting harder to find? *American Economic Review*, 110(4):1104–1144.

Chow, T., Halperin, B., and Mazlish, J. Z. (2026). Transformative AI, existential risk, and real interest rates. Working paper; first posted January 2023.

Coyle, D. and Poquiz, J. L. S. (2025). Making AI count: The next measurement frontier. NBER Working Paper 34330, National Bureau of Economic Research.

Davidson, T., Halperin, B., Houlden, T., and Korinek, A. (2026). When does automating AI research produce explosive growth? Feedback loops in innovation networks. NBER Working Paper 35155, National Bureau of Economic Research.

Demirer, M., Fradkin, A., Tadelis, N., and Peng, S. (2025). The emerging market for intelligence: Pricing, supply, and demand for LLMs. NBER Working Paper 34608, National Bureau of Economic Research.

Demirer, M., Musolff, L., and Yang, L. (2026). Writing code vs. shipping code: Productivity effects across generations of AI coding tools. NBER Working Paper 35275, National Bureau of Economic Research. Revised September 2026.

Duffy, J. and Papageorgiou, C. (2000). A cross-country empirical investigation of the aggregate production function specification. *Journal of Economic Growth*, 5(1):87–120.

Dyatlov, S. (2018). Notes on hyperbolic dynamics.

Erdil, E. and Besiroglu, T. (2022). Algorithmic progress in computer vision. *arXiv preprint arXiv:2212.05153*.

Erdil, E., Potlogea, A., Besiroglu, T., Roldan, E., Ho, A., Sevilla, J., Barnett, M., Vrzla, M., and Sandler, R. (2025). GATE: An integrated assessment model for AI automation. arXiv:2503.04941.

Farboodi, M., Koh, A. J., and Xia, A. (2026). Data-driven automation. NBER Working Paper 35320, National Bureau of Economic Research.

Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2025). Penn world table version 11.0. Groningen Growth and Development Centre.

Feldman, W. M. (2021). Stable manifold theorem. MATH 6410: Ordinary Differential Equations, University of Utah. Fall lecture slides.

Fonteneau, F., Mollins, J., Marchi, S., Russo, L., Gentaz, A., Daoud, M., and André, A.-A. (2025). Advancing the measurement of investments in artificial intelligence. Technical Report 47, OECD Publishing, Paris.

Gans, J. S. (2024). Market power in artificial intelligence. NBER Working Paper 32270, National Bureau of Economic Research.

Gans, J. S. (2025). Growth in AI knowledge. NBER Working Paper 33907, National Bureau of Economic Research.

Hernandez, D. and Brown, T. B. (2020). Measuring the algorithmic efficiency of neural networks. *arXiv preprint arXiv:2005.04305*.

Jones, B. F. (2025). Artificial intelligence in research and development. NBER Working Paper 34312, National Bureau of Economic Research.

Jones, C. I. (1995). R&D-based models of economic growth. *Journal of Political Economy*, 103(4):759–784.

Jones, C. I. and Tonetti, C. (2026). Past automation and future A.I.: How weak links tame the growth explosion. Working paper, Version 0.5, May 2026.

Jones, L. E. and Manuelli, R. E. (1990). A convex model of equilibrium growth: Theory and policy implications. *Journal of Political Economy*, 98(5, Part 1):1008–1038.

Karabarbounis, L. and Neiman, B. (2014). The global decline of the labor share. *Quarterly Journal of Economics*, 129(1):61–103.

Korinek, A., Jones, C. I., Sacher, S., Cotter, T., and McCrory, P. (2026). Economic scenarios for transformative AI. Working Paper 2026-02, The Anthropic Institute.

Korinek, A. and Lockwood, L. M. (2026). Public finance in the age of AI: A primer. NBER Working Paper 34873, National Bureau of Economic Research.

Korinek, A. and McKelvey, P. (2026). Measuring the AI economy. Staff Working Paper 2026-20, Bank of Canada.

Korinek, A. and Suh, D. (2024). Scenarios for the transition to AGI. NBER Working Paper 32255, National Bureau of Economic Research.

Korinek, A. and Vipra, J. (2025). Concentrating intelligence: Scaling and market structure in artificial intelligence. *Economic Policy*, 40(121):225–256.

Melina, G. and Villa, S. (2025). From servers to rates: AI, ICT capital, and the natural rate. IMF Working Paper 25/224, International Monetary Fund.

Mullainathan, S. and Rambachan, A. (2025). Science in the age of algorithms. In Agrawal, A. K., Brynjolfsson, E., and Korinek, A., editors, *The Economics of Transformative AI*, chapter 13. University of Chicago Press.

Nordhaus, W. D. (2021). Are we approaching an economic singularity? Information technology and the future of economic growth. *American Economic Journal: Macroeconomics*, 13(1):299–332.

OECD (2025). OECD Global Long-Run Economic Scenarios: 2025 Update. Technical Report 36, OECD Publishing, Paris.

OECD (2026). OECD Compendium of Productivity Indicators 2026. Technical report, OECD Publishing, Paris.

Pitchford, J. D. (1960). Growth and the elasticity of factor substitution. *Economic Record*, 36(76):491–504.

Rachel, L. (2025). What next for r*? A capital market equilibrium perspective on the natural rate of interest. *Brookings Papers on Economic Activity*, pages 131–187.

Ray, D. and Mookherjee, D. (2022). Growth, automation, and the long-run share of labor. *Review of Economic Dynamics*, 46:1–26.

Rebelo, S. (1991). Long-run policy analysis and long-run growth. *Journal of Political Economy*, 99(3):500–521.

Restrepo, P. (2025). We won't be missed: Work and growth in the AGI world. NBER Working Paper 34423, National Bureau of Economic Research.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5, Part 2):S71–S102.

Sastry, G., Heim, L., Belfield, H., et al. (2024). Computing power and the governance of artificial intelligence. *arXiv preprint arXiv:2402.08797*.

Trammell, P. and Korinek, A. (2023). Economic growth under transformative AI. NBER Working Paper 31815, National Bureau of Economic Research. Revised April 2026.

United Nations (2024). World Population Prospects 2024: Summary of Results. Technical Report UN DESA/POP/2024/TR/NO. 9, United Nations, Department of Economic and Social Affairs, Population Division, New York.

Wachter, J. and Wachter, J. (2026). What investment data implies about the AI transition. NBER Working Paper 35290, National Bureau of Economic Research.